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RESEARCH ARTICLE

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Designing assessments in the generative AI era: A tailored assessment framework for ICT tertiary education

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Abstract

The rapid expansion of generative artificial intelligence (GenAI) as a disruptive force in modern times presents tertiary education with both unprecedented opportunities and profound challenges for assessment design. As GenAI can create plausible assessment responses with limited student engagement, many traditional assessments are now at risk of becoming obsolete. Assessment reform is particularly important in ICT programs, where GenAI can readily generate assessment content, such as coding, and algorithms, and technologically savvy students may be prone to misuse it rather than engage in authentic learning. Consequently, sector-wide discussions are centred on how to redesign assessment to enable the appropriate and ethical use of GenAI while ensuring students genuinely achieve the expected learning outcomes. Despite calls for updated strategies from quality assurance bodies, a significant gap remains between the pace of GenAI advancement and the implementation of such assessment reforms. The objective of this study is to identify how educators can design assessments for ICT tertiary students that accurately measure student learning outcomes in an era of widespread access to GenAI tools. Using a qualitative interpretivist approach and thematic analysis of scholarly and institutional practice documents, this research proposes the GENESIS framework (GenAI-Enabled Strategic and Instructional System for Assessment), a five-pillar, institution-wide model emphasising shifts in institutional values, governance, and support as prerequisites for sustainable assessment reform. GENESIS is complemented by the Assessment Design Matrix (ADM), a tripartite structure integrating graduate competencies, assessment strategies, and assessment methods. ADM provides a practical, hands-on guide for assessment designers to craft integrity-focused assessments that assess ICT graduate competencies. Together, GENESIS and ADM propose a structured, discipline-tailored framework to ensure assessments continue to validly measure student learning within the GenAI ecosystem.

Highlights

- GENESIS is a five-pillar, institution-wide practice framework that embraces changes in holistic values, governance, and support systems, forming a sustainable foundation for assessment design suited to the GenAI era.

- The proposed targeted assessment design via ADM introduces a matrix linking graduate competencies, assessment strategies, and assessment methods, demonstrating how competencies inform strategy selection, strategies guide method choice, and methods generate valid evidence of competency attainment.
- The GENESIS and ADM frameworks provide educators with actionable tools that connect graduate competencies to suitable strategies and methods, producing an industry-aligned and pedagogically robust approach to assessment design.

Keywords Generative AI, ICT higher education, Assessment design, Assessment frameworks, Assessment strategies, Graduate competencies

Introduction

Since OpenAI's release of GPT-3 in 2020, followed by DALL·E and CLIP, large language models (LLMs) have progressed rapidly. The launch of ChatGPT in late 2022 and GPT-4 in 2023 brought GenAI into mainstream use, offering advanced reasoning and multimodal capabilities (OpenAI, 2023). By 2025, GenAI has become deeply integrated into the academic practices of both students and educators. This transformative rise of GenAI has created a pressing need to rethink assessment design in tertiary education. It's widespread adoption, triggered concerns over academic integrity (Cotton et al. 2024), intellectual property, privacy, and equity (McDonald et al. 2025). Institutional responses ranged from proactive adoption to cautious resistance, reflecting ongoing uncertainty about GenAI's role in assessment design (Nikolic et al. 2024).

In Australia, this ambiguity began to resolve when the Tertiary Education Quality and Standards Agency (TEQSA) issued a national directive in June 2024 requiring all higher education providers to submit GenAI integration plans. In response, most universities and higher education providers adopted GenAI-permissive policies, while TEQSA released a national implementation toolkit, establishing a foundation for more unified, pedagogically sound GenAI assessment practices (TEQSA, 2024). Despite ongoing efforts by higher education providers to revise assessment strategies in response to GenAI, a significant gap remains between the pace of technological advancement and assessment implementation. The rapid evolution of GenAI has outpaced the development of effective integration frameworks for academic assessment (Kasneci et al. 2023). Recent assessment frameworks and guidelines proposed within the sector have introduced several promising directions such as the traffic light systems (Corbin et al. 2025), the two-lane approach (Curtis, 2025), and the AI Assessment Scale (AIAS) (Perkins et al. 2024). While these frameworks offer valuable conceptual guidance, they remain broad, lack discipline-specific applicability, and have yet to be validated in authentic educational settings. Many focus primarily on communicating institutional rules rather than addressing the deeper redesign of assessment mechanisms. Consequently, despite these emerging proposals, significant challenges persist in translating such strategies into practical, scalable, and discipline-relevant assessment solutions.

Building on the research gap that discipline specific assessment design requires further investigation, this study focuses on assessment reform tailored to the ICT discipline, recognising the need for frameworks that address specific student cohorts and disciplinary contexts in the GenAI era. Core ICT tasks, such as coding, algorithm development, and system design, can now be completed rapidly using GenAI with minimal student effort, placing traditional assessments at risk of failing to measure genuine learning outcomes. Moreover, ICT students' advanced technical skills may increase the likelihood of GenAI misuse, further emphasising the need for integrity focused, discipline specific assessment redesign. This study targets ICT higher education students at AQF levels 7 to 9 (AQF, 2013) a cohort typically specialising in areas such as cybersecurity, software development, data analytics, information systems, and machine learning. ICT tertiary students' graduate competencies focus on developing the workforce skills needed to use and manage digital technologies, including computing, software, networks, and information systems, across diverse industries. To succeed, ICT graduates must also develop core skills in problem-solving, abstraction, design, ethics, professionalism, teamwork, and communication (ACS, 2015). Given the distinct learning outcomes and technical demands of ICT domain, this study addresses the research question: *What assessment strategies, techniques, and methods are most appropriate for ICT tertiary assessments in the context of GenAI?* The objective of this study is to identify how educators can design assessments for ICT tertiary students that accurately measure student learning outcomes in an era of widespread access to GenAI tools.

Since late 2022, academics and academic institutions have published a growing body of GenAI-related resources, including scholarly articles, policy documents, practice guides, and web-based materials such as concept papers. While these sources offer valuable insights, much of the informal documentation remains under-analysed. This study aims to synthesise this emerging body of knowledge, spanning assessment models, frameworks, and academic practices, to develop a discipline specific framework tailored to ICT assessment in higher education in Australia. In doing so, it contributes to the broader discourse on GenAI-enabled assessment reform, offering practical recommendations to strengthen academic integrity and improve learning outcomes. Adopting a qualitative interpretivist approach, the study employs thematic analysis to examine relevant secondary documents and extract key strategies responsive to the evolving educational landscape.

Methodology and analysis

Methods

This study adopts a web-based document analysis approach, supported by thematic analysis to identify and interpret patterns in the data. Bowen (2009) defines document analysis as "a systematic procedure for reviewing or evaluating documents, both printed and electronic material" (p. 27). Although often used alongside other methods in mixed-method research and triangulation, document analysis can also be effective as a standalone method (Morgan, 2022; Scarparolo and Porta, 2025). The widespread availability and accessibility of online content present significant opportunities for research using web-based document analysis. Several recent studies have successfully employed this approach using only publicly available online sources (Chen et al. 2021; Perkins & Roe, 2024). In line with these works, this study focuses on analysing publicly accessible web-based documents to explore themes related to GenAI-enabled assessment practices in ICT higher education. While relying on publicly available, predominantly English-language sources may limit the breadth of evidence, this was a deliberate choice to ensure materials were verifiable, consistently accessible, and comparable across institutions. Publicly available documents also reflect the official positions, policies, and strategic intentions that institutions communicate to students, regulators, and the broader public, making them highly relevant to an analysis concerned with sector-level transparency and assessment design. The potential limitations of restricted access or linguistic bias are acknowledged, but the method remains appropriate for the study's aims.

Thematic Analysis (TA) was chosen as a suitable method for this purpose. TA involves identifying patterns or themes within qualitative data (Braun and Clarke, 2006; Clarke and Braun, 2013) and supports a flexible, theory-informed exploration of meanings, particularly in unstructured data. Its broad application in qualitative research reinforces its appropriateness. Several studies have demonstrated TA's effectiveness as a standalone method for analysing web-based documents (Izadi et al. 2023; Orr et al. 2024; Sarshar et al. 2021) In this study, TA was applied to systematically identify, analyse, and interpret themes related to assessment reforms (Clarke and Braun, 2017). As TA does not require adherence to a specific theoretical framework, it aligns well with an interpretivist approach (Braun and Clarke, 2006). This study follows the TA process outlined by Braun and Clarke (2006), which includes a six-stage process: (1) data familiarisation, (2) code generation, (3) theme development, (4) theme review, (5) theme definition, and (6) report production.

Data set and search strategy

Braun and Clarke (2006) define a data set for thematic analysis as the portion of the data corpus selected for a specific analysis (p. 79). To examine academic perspectives on evolving assessment practices in response to GenAI, this study analyses a range of online documents published by scholars and practitioners over the past two years. The dataset comprises formal and informal publicly available web-based resources, including journal articles, university policies and procedures, and practice and concept papers published between 2023 and 2025. All selected documents relate to assessment reform in the context of GenAI within tertiary-level ICT qualifications. Due to the researcher's language proficiency, only English-language documents were included.

Drawing on recent studies employing web-based document analysis with thematic analysis (Alexander, 2022; Almomani et al. 2023; Chen et al. 2021; Kallio et al. 2020; Perkins and Roe, 2024), this study adopted a structured search strategy. Using Google via the Chrome browser over five days (March 5–9, 2025), the researcher applied the following search phrases: *"assessment strategies AND Generative AI," "computer science assessments AND generative AI," "information technology assessments AND generative AI,"* and *"ICT assessments AND ChatGPT."* Documents were selected based on their order of appearance in the search results. The collected materials were primarily in PDF format or hosted on web pages. To ensure objectivity and avoid conflicts of interest, sponsored links were disregarded. Because the study examines GenAI-enabled assessment within tertiary settings, documents from primary or secondary education were excluded unless they provided content with direct applicability to higher education (e.g., cross-sector AI policies, assessment standards, or government directives that span multiple education levels). Additionally, papers lacking academic rigour, quality, or professional affiliation were omitted from the dataset. Ethical approval was not required, as all data were obtained from publicly accessible sources. The final dataset comprises 80 documents, detailed in Appendix 1.

Analysis process

Qualitative data analysis of the selected web-based documents was conducted, following Braun and Clarke's (2006) six-phase thematic analysis framework. In the first phase, the researcher selected the dataset and engaged in repeated, in-depth readings throughout March 2025 to gain a comprehensive understanding. This immersion reflects Braun and Clarke's (2006) view that familiarisation forms "the bedrock for the rest of the analysis" (p. 87).

In the second phase, initial codes were generated by organising the content into meaningful themes and sub themes using Delve software. The researcher systematically reviewed each document, giving equal attention to all data while avoiding the influence of pre-existing theoretical frameworks. Codes were assigned whenever relevant or noteworthy data aligned with the study's research focus. This process was guided by specific research questions, with the aim of assigning a code to each pertinent feature (Braun and Clarke, 2006). Determining when to stop coding in an interpretive study is not always clear-cut. However, applying the principle of data saturation from grounded theory, coding continued until no new themes or subthemes emerged (Glaser and Strauss, 2017). Themes were considered well-developed when they were rich, nuanced, and supported by sufficient data extracts. Saturation was reached after analysing 72 documents; eight additional documents were reviewed to confirm saturation, completing the open coding process for all 80 sources. This process, conducted from April to May 2025, resulted in 185 initial codes. Appendix 2 lists the derived codes.

The third phase began on June 1, 2025. This stage involved grouping codes into broader themes and organising related data extracts under each. The analyst examined relationships among codes to identify overarching patterns using Delve software. The Edraw.ai collaboration tool was used to help visualise connections among codes, subthemes, and themes, supporting the development of the preliminary thematic map. In the fourth phase, the analyst reviewed and refined emerging themes through two levels of evaluation: first, assessing internal coherence of coded extracts within each theme; second, evaluating how well the thematic structure reflected the entire dataset. The thematic map was also reviewed to ensure it accurately captured the dataset's broader meanings (Braun and Clarke, 2006).

By June 20, 2025, the fifth phase commenced focusing on defining and refining the themes and presents the findings in a thematic map. Edraw.ai's intuitive interface facilitated visual organisation and analysis of the themes to review the visual thematic map. The analyst reviewed each theme in depth, clarified its core meaning, and examined supporting data extracts. Refinement continued until theoretical saturation was achieved, indicating a shift from description to interpretation and a comprehensive understanding of key concepts and relationships. At this point, the research question was deemed fully addressed (Glaser and Strauss, 2017). Figure 1 presents the final thematic map, and Table 1 defines each theme.

In the sixth and final phase, the author compiled a comprehensive report of the findings, presented in the subsequent sections of this paper. During the preparation of this report, the author utilised two large language models (LLMs), ChatGPT (GPT-4 Omni) and Gemini 2.5 Flash, for language and copy-editing support. To enhance reliability, both tools were used to cross-check the manuscript and improve its overall formatting. Their use was strictly limited to refining grammar, clarity, and conciseness during the

Fig. 1 Thematic map

drafting and revision stages. No content was completely generated or interpreted by the LLMs. The author retains full responsibility for the originality, accuracy, and integrity of the content presented.

Results

The thematic analysis yielded 7 key themes and 70 subthemes that collectively address the research question. The findings indicate that adapting ICT assessment in response to GenAI necessitates a whole-of-institution approach. This transformation is structured around seven core themes: *Values and Dispositions* (T1), *Governance* (T2), *Support Infrastructure* (T3), *AI-Integrated Curriculum* (T4), *Graduate Competencies* (T5), *Assessment Strategies* (T6), and *Assessment Methods* (T7). Notably, the analysis revealed that these themes are interconnected in a sequential pattern, suggesting an order in which the transformation should unfold, beginning with T1 and culminating in T7. This progression implies that addressing foundational values and dispositions is essential before implementing specific assessment methods, with the remaining themes ideally addressed in the proposed sequence. The thematic map presented in Fig. 1 visually illustrates how these themes represent a comprehensive transformation of ICT assessment practices. Table 1 presents definitions of the key themes and their associated subthemes.

The following subsections present the findings related to the seven key themes that emerged from this study. The specific excerpts taken from the analysed documents are denoted as S# followed by the document number.

Table 1 Emerged themes from thematic analysis

Key themes and definitions	Sub themes
Values & Dispositions (T1) The core belief system of an educational institution that guides attitudes, decisions, and interactions. It shapes how learners and educators approach teaching, learning, collaboration, and professional responsibility in the context of GenAI use	Cultivate Innovation Culture (VD1) Adopt a Facilitative Approach Instead of Policing (VD2) Integrity, Collaboration & Responsibility (VD3) Behavioural and Attitude Changes (VD4)
Governance (T2) A comprehensive institutional framework for regulating the use of GenAI in learning and assessment, encompassing academic integrity policy alignment, responsible use guidelines, proactive misconduct detection, equitable access, student support, and the management of emerging risks to privacy, and security	Academic Integrity & GenAI Policy Reform (G1) Quantification of AI allowance (G2) Academic Misconduct Detection Revision (G3) Special Considerations Policy Revision (G4) Ethical & Equitable Digital Transformation (G5) Assessment Privacy, Security & Risk Management Processes (G6)
Support Infrastructure (T3) A comprehensive, institution-wide supportive infrastructure that ensures equitable access to GenAI tools, builds digital and assessment literacy, supports academic and emotional well-being, reduces workload through automation, and promotes clear communication of policies across the academic community	Upskill Academics (SI1) Automated Assessment Systems (SI2) Equity and Access for Resources (SI3) Driving Student Motivation (SI4) Emotional Drivers (SI5) AI-Enabled Tutoring Support (SI6) Communication & Expectation Clarity (SI7)
AI-Integrated Curriculum (T4) The structured integration of AI-related content and competencies into academic programs to align learning outcomes with evolving industry needs	Programme Level Assessment Alignment (AC1) Recalibrating Assessment Volume (AC2) GenAI Literacy as a Graduate Attribute (AC3) Quality Gatekeeping Assessments (AC4) Assessment Triangulation (AC5)
Graduate Competencies (T5) The holistic combination of knowledge, skills, attitudes, and ethical awareness that enable graduates to adapt, collaborate, communicate, and lead effectively in complex and evolving professional and academic environments	Prompt Engineering (GC1) Metacognitive Skills (GC2) Linguistic skills (GC3) Emotional intelligence (GC4) Decision-making, & Leadership (GC5) Teamwork & Conflict Resolution (GC6) Ethical Intelligence (GC7) GenAI Literacy & Strategic Use (GC8) Communication & Confidence (GC9) Academic Writing and Referencing (GC10)
Assessment Strategies (T6) Deliberate and structured approaches to design and implement assessments in order to achieve specific learning outcomes	Assess Foundational Knowledge in Person (AS1) Use GenAI to Tutor Low-Stakes Assessments 24/7 (AS2) Ungrading for Low-Stake Assessments (AS3) Conduct Low-Stakes Assessments Frequently (AS4) Implement Multi-staged, Connected Scaffolding (AS5) GenAI Assisted Assessment Drafting & Planning (AS6) Assess Process over Product (AS7) Assess Higher Order Thinking Over Theoretical Recall (AS8) Personalise Assessments with Authentic Contexts (AS9) Facilitate Collaboration with GenAI (AS10) Implement Flipped AI-Driven Assessments (AS11) Restrict Assessment Open Time (AS12) Facilitate a Continuous Feedback Loop (AS13) Integrate Multimodal Artefacts Innovatively (AS14)

Table 1 (continued)

Key themes and definitions	Sub themes
	GenAI De-Scaffolding (AS15)
	Include Essential Critical Thinking Aspects in Assessments (AS16)
	Teachers use GenAI for Assessment Diagnostics (AS17)
	Enforce Supervision / invigilation for Gatekeeping Assessments (AS18)
Assessment Methods (T7) Tools or techniques used to collect evidence to evaluate whether students have achieved the expected learning outcomes	Process Documentation with Iterative Evidence (AM1)
	Annotated Bibliographies (AM2)
	Debugging or Code Tracing Tasks (AM3)
	Hackathon-Style Challenges (AM4)
	Interactive Hands-on Tutorials & labs (AM5)
	Competency Based Assessments (AM6)
	Controlled Exams & quizzes (AM7)
	In class Concept Maps & Story Telling or Flowchart Design (AM8)
	Peer Reviews (AM9)
	Self-Reflections, Portfolio & Journals (AM10)
	Experiential Project Based Learning (AM11)
	Authentic or Context based assessments (AM12)
	Simulations & Role-Plays (AM13)
	Milestone Based Group Projects (AM14)
	Interviews, Presentations or Demonstrations (AM15)
	Real-life Problem-Solving Assessments (AM16)
	Multimodal Artifacts (AM17)
	Capstone Projects and Professional Experience (AM18)
	Multi-Staged-Connected Essays or Written Reports (AM19)

Theme 1—Values & Dispositions (T1)

Theme 1 shows that meaningful assessment transformation in response to GenAI begins with institutional values rather than technical solutions. Before implementing policy or operational changes, institutions need a coherent set of shared beliefs that shape how staff and students approach teaching, assessment, integrity, and professional responsibility in the GenAI era. These values are reflected across four interrelated dispositions (VD1–VD4).

The data highlight the importance of cultivating an innovation-oriented culture (VD1). Institutions must encourage creativity, experimentation, and a willingness to test new approaches. Several sources emphasise viewing GenAI experimentation as an opportunity for improvement, *"*Seeing GenAI experimentation as a way to improve professional and educational practices can encourage a culture of adaptation and inquiry.*"* (S#39) and treating mistakes as learning rather than failure. This mindset enables institutions to adapt to rapid technological change and prevents assessment reform from becoming reactive or stagnant. As one source notes, leaders should "*foster a culture of innovation and continuous improvement*" (S#42). The findings also show that institutions need to adopt a facilitative rather than policing orientation toward GenAI (VD2). While it remains clear that *"submitting AI-generated content as your own work is considered academic misconduct"* (S#80), over-reliance on detection tools has led to inequitable outcomes. The findings highlight challenges from the ongoing 'cat-and-mouse' dynamic between students and academics. Non-native English speakers are particularly disadvantaged when AI-detection tools misclassify their writing (S#28), and some students respond by running their work through multiple paraphrasing tools to avoid false positives (S#19). These dynamics reinforce that sustainable integrity must be built through education and guidance, not punitive surveillance. As one dataset notes, "retributive justice fosters an adversarial us-versus-them climate" (S#75), highlighting the need for restorative, supportive practices.

Integrity, collaboration, and shared responsibility (VD3) further underpin assessment transformation. The data consistently call for building communities that uphold ethical AI use together, with *"both faculty and students developing an understanding of the ethical use of AI"* (S#08). Encouraging students to champion integrity among peers (S#24) and enabling open reporting of concerns (S#21) strengthens collective accountability. These values also align with preparing ICT students for professional practice, where responsible GenAI use is an essential competency rather than an optional skill (S#75). Finally, successful integration of GenAI requires behavioural and attitudinal shifts (VD4). Resistance to change remains a significant barrier (S#43), underscoring the need to reshape underlying beliefs about academic integrity and ethical scholarship. As one source notes, *"academic integrity is about shared values, norms and conventions… equipping learners with the capabilities and values necessary to conduct ethical scholarship"* (S#07). The evidence shows that cultivating ethical awareness and social responsibility (S#20) is essential for embedding GenAI use within everyday academic practice.

Taken together, VD1–VD4 reinforce that assessment innovation is fundamentally a values-driven process. Institutions must move beyond simply adopting new tools and instead develop the cultural conditions (e.g. innovation, facilitation, shared responsibility, and ethical mindsets) needed for sustainable transformation. Figure 2 summarises the emerged four sub-themes emerged under key theme 1.

Theme 2—Governance (T2)

Once foundational values are established, institutions must embed them into governance structures that enable sustainable assessment transformation. The data show that governance needs to move beyond informal practices toward clear, formalised policies, frameworks, and decision-making processes that regulate GenAI use in learning and assessment (S#16; S#42). Thus, *Governance* emerges as the next key theme: a comprehensive framework regulating GenAI use in learning and assessment, including integrity policies, responsible use guidelines, misconduct detection, equitable access, student support, and risk management.

The first sub-theme, Academic Integrity and GenAI Policy Reform (G1), stresses the urgent need for institutions to formally regulate GenAI in learning and assessment by revising Academic Integrity Policies to clarify the institution's stance (S#42; S#33; S#80). Many providers have begun introducing models such as "traffic light" or "two-lane" systems, yet staff report that implementation remains inconsistent and confusing (S#75; S#42). As one source cautions, *"unclear or poorly thought-out policies will only serve to confuse students and make enforcing policies difficult"* (S#75). The evidence underscores that effective governance requires institution-wide policy development involving educators, students, and industry experts to ensure relevance and legitimacy (S#37; S#18; S#80). The sub-theme Quantification of AI Allowance (G2) postulate that the institutions must clearly define the permissible extent of GenAI use across different assessment types and levels. Policies vary, some use percentage-based limits, while others rely on qualitative descriptions or hybrid approaches (S#10; S#14). Regardless of format, the key message is that institutions must clearly articulate how much GenAI use is acceptable

Fig. 2 Sub-themes emerged from the Key Theme T1

in different assessment contexts so that expectations align with pedagogical intent and long-term educational goals.

Academic Misconduct Detection Revision (G3) sub-theme highlights the need to update misconduct policies reflecting GenAI's challenges. Early reliance on traditional tools such as Turnitin proved problematic due to false positives, bias, and inconsistent accuracy (S#41; S#19; S#45). This study finds significant concerns over accuracy, bias, and fairness of current tools (S#41; S#19; S#45; S#7; S#50), urging a shift to process-oriented approaches emphasising in-person assessment and evidence collection (S#14; S#63; S#29; S#24). As one source notes, *"by redesigning our assignments to include in-class elements... we can make GenAI use easier to detect and mitigate"* (S#29). Poor assessment design contributes to misconduct (S#32; S#19), highlighting the need for governance to support assessment redesign that measures higher-order thinking (S#14; S#12; S#10). Special Considerations Policy Revision (G4) addresses evolving criteria for special considerations due to GenAI-related disruptions, such as shorter extensions encouraging GenAI reliance or mandated AI platforms causing access issues (S#31; S#35). Short extensions may incentivise GenAI overuse, and mandated use of specific AI tools can create legitimate access concerns. As one data point notes, *"the new approach is an opportunity to rethink special consideration, including simple extension"* (S#06). Institutions must ensure that updated policies maintain academic integrity while supporting student wellbeing in an AI-driven environment (S#09; S#42).

Ethical & Equitable Digital Transformation (G5) stresses institutional responsibility for commit to ethical and equitable digital transformation for the entire academic community. Policies must safeguard informed consent (S#67; S#45), protect student intellectual property, *"a student's work is their intellectual property"* (S#36), and address cultural bias and fairness concerns (S#68; S#28). Equitable access remains a significant challenge, with some students lacking digital literacy or reliable internet (S#43; S#34). While some institutions provide AI tool subscriptions to ensure equal access (S#42; S#10), disparities continue to affect assessment fairness. Another key theme identified in the data analysis is the emerging concerns regarding, Assessment Privacy, Security, and Risk Management (G6), which addresses risks related to personal data privacy in GenAI tool use. GenAI tools introduce new vulnerabilities, including inappropriate handling of personal data, accidental uploading of confidential materials, and the risk of AI hallucinations affecting grading accuracy (S#19; S#28; S#68). One document emphasises that *"failure to adequately consider privacy issues... could have legal consequences"* (S#77). Governance must therefore ensure compliance through Privacy Impact Assessments, secure data handling protocols, and alignment of AI adoption with pedagogy and infrastructure (S#30; S#07; S#04).

Overall, Theme 2 demonstrates that effective governance is essential for coherent, fair, and ethically grounded GenAI integration. Figure 3 summarises the six sub-themes, illustrating how policy clarity, integrity protections, equitable access, and robust data governance collectively shape the institutional capacity to navigate GenAI-enabled assessment transformation.

Theme 3—Support infrastructure (T3)

Theme 3, *Support Infrastructure* (T3), highlights the institution-wide systems required to enable ethical, equitable and effective GenAI use. Across the data, this theme reflects the need for accessible tools, strengthened digital and assessment literacy, reduced administrative burden through automation, and clear, consistent communication across the academic community.

The first sub-theme under *Support Infrastructure* (T3), Upskilling Academics (SI1), stresses the need to build GenAI literacy and assessment capability among staff through training and mindset change (S#70; S#59; S#34). Several statements note that educators are not necessarily more proficient than students related to GenAI use (S#75; S#66), reinforcing the need for mindset change, discipline-wide training access, and ongoing support. For example, educators reported uncertainty about integrating tools such as ChatGPT into teaching and guiding students' appropriate use (S#66), indicating that capability-building is foundational to any GenAI-enabled assessment reform. The second sub-theme identified in the data is the use of Automated Assessment Systems (SI2) to support academic processes by reducing workload and supporting timely feedback. Tools such as OpenAI Codex and web-based prototypes (S#77; S#74) were seen to streamline routine marking, particularly for free-text responses, a traditionally time-intensive task. Instant formative feedback (S#58; S#77; S#11) enables personalised learning and supports educators in scaling feedback (S#18; S#20; S#15). However, concerns around accuracy, reliability, and opaque decision-making (S#38; S#68; S#54) indicate that institutions must evaluate pedagogical soundness before widespread adoption. Sub-theme Equity and Access to Resources (SI3) emerged as a critical governance requirement (aligned with G5), emphasising that students should have equal access to GenAI tools, training, and literacy-building activities regardless of background or discipline (S#42; S#24). One statement notes the need for structured training to build confidence in AI-supported assessment (S#45), reinforcing that institutions must create inclusive, digitally capable learning environments to prevent widening achievement gaps.

Fig. 3 Sub-themes emerged from the Key Theme T2

Driving Student Motivation (SI4) is another sub-theme that emerged under the theme T3, which further reinforces the need for supportive, trust-based environments that encourage deep learning rather than surveillance-driven behaviours. Intrinsic motivation can be nurtured through initiatives such as mentoring programs, group-based learning sessions, and personalised academic counselling, all of which help students find personal relevance and meaning in their studies (S#04; S#54; S#51; S#11). Extrinsic motivators, such as recognising ethical use and creative application of GenAI (S#51), were seen to help redirect focus from short-term grades to long-term growth. Building on the above discussion of motivational drivers, the study also identifies Emotional Drivers (SI5) as a critical sub-theme. Students reported anxiety, perceived bias, loss of control, and tool fatigue when interacting with GenAI systems (S#04; S#01; S#18; S#07). For instance, neurodivergent students may find tool interfaces overwhelming or ambiguous (S#01), and evolving assessment formats (e.g. in-class Q&A) pose challenges for those with social anxiety (S#27; S#25). The findings reaffirm the central role of human educators in providing tailored emotional support, maintaining motivation, and guiding students' engagement (S#04). Institutions should therefore design inclusive emotional support structures, ensuring accessibility for diverse and multilingual cohorts, for example, through translation support or multimodal tools like speech-to-text (S#80).

AI-Enabled Tutoring Support (SI6) was frequently described as a scalable and affordable mechanism for academic assistance, particularly in resource-constrained settings (S#59). GenAI tools were seen to democratise access to guidance, mirroring the way early internet access once widened opportunity gaps (S#59). Examples include using AI as a "study buddy" or for generating personalised study guides (S#29; S#70). Institutions can leverage these systems to strengthen out-of-class support, enhance tutorials, and assist with career development. Communication and Expectation Clarity (SI7) is the next sub-theme that emerged from the findings. Several statements highlighted that communication regarding GenAI related policies and procedures has been insufficient, leading to notable gaps in understanding within the academic community (S#52; S#38). Strengthening communication was viewed as essential once governance reforms are established, including strategies such as regular updates, tailored messages for specific cohorts, dedicated FAQs, and AI-powered chatbots to address common queries (S#42; S#19). Clear and consistent communication ensures that all members of the academic community understand their responsibilities and the institution's expectations.

Figure 4 summarises these sub-themes, illustrating the elements necessary to develop an institution-wide support system that enables a well-resourced, inclusive, and ethically responsible environment for effective GenAI use.

Theme 4—AI-integrated curriculum (T4)

AI-Integrated Curriculum (T4) emphasises the need to formally embed GenAI considerations into curriculum design. This revision must also involve a re-evaluation of Program Learning Outcomes (PLOs) to align with evolving industry expectations and the competencies required of future graduates (S#80; S#24; S#16; S#12). This theme serves as the transition point through which earlier institutional shifts, values (T1), governance (T2), and support infrastructure (T3), become operationalised within teaching and learning. Overall, the data highlight that curricular transformation must uphold academic rigour, ensure workload balance, support diverse skills, and remain responsive to broader societal needs.

The first sub-theme emerging from this theme is Programme-Level Assessment Alignment (AC1). This sub-theme underscores the need for a coordinated and systematic approach to the design and periodic review of assessments across the entire academic program. Respondents consistently emphasised that assessment reform in the GenAI era cannot rely on isolated, unit-level decisions, but requires a coordinated, program-wide approach. This ensures that Course Learning Outcomes (CLOs), Program Learning Outcomes (PLOs), and graduate attributes evolve coherently and reflect the capabilities expected by industry. As one document noted, *"assessment design considerations should span a whole program… rather than be applied solely at individual task or unit levels"* (S#16). Another highlighted that when module leads design assessments independently, students experience inconsistent expectations and fail to build skills progressively (S#25). These insights underline that constructive alignment (i.e. mapping assessments systematically to program-level outcomes) is essential for eliminating redundancy, ensuring consistency, and maintaining a clear focus on what students are expected to learn (S#29).

Another key consideration is the need for Recalibrating Assessment Volume (AC2) during the transformation toward an AI-Integrated Curriculum. As assessments increasingly prioritise higher-order thinking and as in-class, staged, or supervised tasks become more common (S#01; S#12; S#43), both student and staff workload must be reconsidered. Findings of this study pointed out the importance of timing assessments outside peak periods, adjusting deadlines, and embedding in-class working time (S#19). The broader implication is that quality, rather than quantity, must drive assessment design, ensuring sustainability and meaningful learning. The next sub-theme emerging from the

Fig. 4 Sub-themes emerged from the Key Theme T3

data is GenAI Literacy as a Graduate Attribute (AC3). The findings underscore the need to integrate GenAI-related competencies into the broader framework of graduate attributes. The data show strong support for embedding GenAI-related competencies across programs, not as optional extras but as essential capacities for future graduates (S#24; S#28). This includes ethical and critical engagement with AI, alongside broader capabilities such as resilience and critical thinking, which help students navigate unpredictable technological futures (S#18). As emphasised by one participant, institutions must balance AI-specific skills with disciplinary knowledge and transferable competencies such as communication and ethical reasoning (S#08). Collectively, the findings suggest that graduate attributes must be updated and explicitly reflected in learning outcomes at both program and unit levels.

Another important sub-theme identified in the findings is the need to incorporate Quality Gatekeeping Assessments (AC4) within the AI-integrated curriculum enforcing assessment security. Respondents noted that securing every assessment task against GenAI is both unrealistic and undesirable; instead, programs should identify critical "stage-gates" (e.g. capstone projects, demonstrations, or milestone evaluations) where heightened assessment security is necessary (S#06; S#07; S#12). Establishing these key checkpoints verify essential knowledge and competencies before students' progress or graduate. As one document explained, this involves "identifying the key assessment moments at a program level and securing those" (S#16). Gatekeeping points also enable earlier detection of learning gaps, supporting timely intervention and ensuring consistent program standards. The final sub-theme emerging from theme T4 is Assessment

Triangulation (AC5). This sub-theme emphasises the need for diverse assessment types across a degree program to ensure fair evaluation of varied student strengths and comprehensive measurement of cognitive, practical, and transferable skills, including those emerging from GenAI use (S#25; S#21). Constructive alignment remains central, ensuring assessments accurately measure learning outcomes (S#36), while triangulation enhances trustworthiness and inclusivity (S#16). Mapping existing assessments to program-level outcomes can also reveal redundancies and gaps, further reduce unnecessary workload and strengthen curriculum coherence (S#24).

Figure 5 synthesises these sub-themes, illustrating how T4 operationalises institutional shifts into a cohesive AI-integrated curriculum. The findings collectively underscore that embedding GenAI into academic programs requires systematic, program-wide curriculum design that maintains integrity, promotes equity, and equips graduates with the competencies needed in a rapidly evolving technological landscape.

Theme 5—Graduate competencies (T5)

The theme emerged from data emphasises the importance of clearly identifying and embedding specific graduate competencies within academic learning and skill development processes. Crucially, these competencies must not only be taught but also systematically assessed to ensure their attainment (S#54; S#48, S#66; S#59; S#57; S#77). The analysis identified the following sub-themes that represent graduate competencies, which must be intentionally embedded, taught, and assessed within ICT degree programs.

Prompt Engineering (GC1) is the ability to design clear, effective prompts for GenAI tools. Several sources emphasised that prompt engineering is now a creative, iterative skill that requires training and assessment (S#55; S#60; S#63). For example, one institution introduced specialised courses in generative AI focusing on prompt engineering (S#55). This competency signals graduates' ability to control and critically shape GenAI

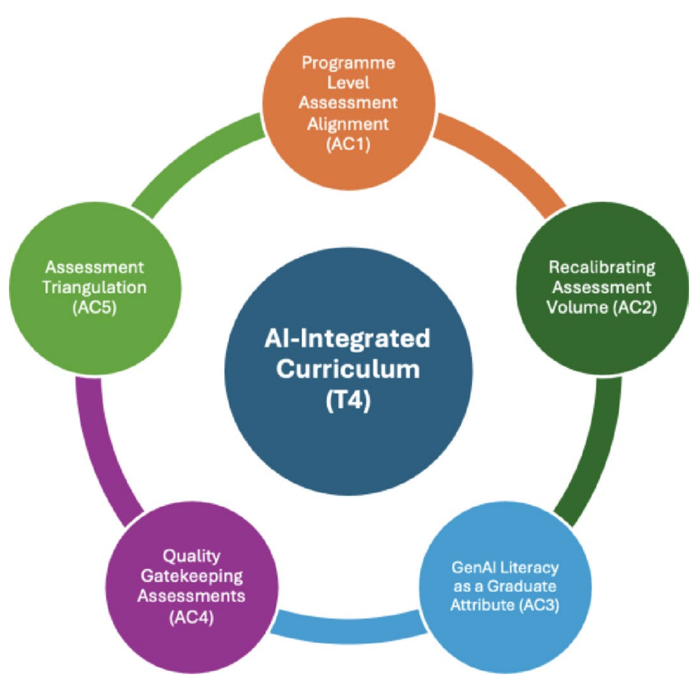

Fig. 5 Sub-themes emerged from the Key Theme T4

outputs. Metacognitive Skills (GC2) refer to students' capacity to monitor, regulate, and reflect on their learning (S#01; S#11; S#25). Finding of the study noted that assessments should capture how students learn (not only what they produce) to promote deeper learning and reflection (S#09; S#07). This competency is essential for developing learners who can adapt to AI-rich environments. Linguistic Skills (GC3) encompass proficiency in reading, writing, speaking, and listening, the skills that are increasingly vital in modern ICT professions where communication, collaboration, and client interaction are essential. The rise of GenAI offers new opportunities to support the development of these skills (S#71; S#37; S#07). GenAI tools can assist students through self-evaluation, idea generation, and language refinement, helping them overcome challenges such as writer's block and enhancing their ability to express complex ideas (S#29; S#80; S#57; S#37). Emotional intelligence (GC4) refers to the capacity to recognise, understand, and manage one's own emotions as well as those of others. Core skills such as problem-solving and emotional competence are foundational and applicable across diverse life and professional contexts. Emotional competence has also emerged as a newly focused intended learning outcome (S#20). Assessment tasks that require leadership, teamwork, and problem-solving under pressure can help develop these competencies (S#18; S#16).

The findings point to several interconnected sub-themes such as GC5, GC6 and GC7 that are vital for graduate capabilities. Decision-Making and Leadership (GC5) reflects the ability to make informed choices and guide others in pursuit of shared goals. Teamwork and Conflict Resolution (GC6) encompass collaboration and constructive dispute resolution, while Ethical Intelligence (GC7) refers to acting with integrity and recognising the ethical implications of decisions. Strengthening student competencies in these areas is essential to ensure they are well-prepared for future professional environments (S#19; S#71). Assessment design should therefore integrate opportunities for students to receive feedback not only on the technical accuracy of their but also on critical thinking, judgment, and ethical reasoning (S#16). Embedding discussions on the limitations and societal implications of emerging GenAI technologies, further supports the development of ethically sound decision-making skills (S#23). Future assessment practices should be designed to evaluate students' capacity to apply logical reasoning, demonstrate creativity in proposing solutions, and leverage data for informed decision-making. Furthermore, these assessments should assess students' abilities to navigate uncertainty, defend their choices, and demonstrate time management as a core component of academic and professional performance (S#71; S#21; S#58; S#01). The findings also highlight group work as an effective strategy to mitigate academic misconduct, as it requires collaborative evidence and mutual agreement on contributions (S#14). However, teaching ethical, moral, and civic values remains a nuanced challenge that depends on human judgment and contextual understanding (S#04). Therefore, essential competencies such as conflict resolution (S#32; S#09) and ethical intelligence (S#43; S#75; S#21) must be intentionally cultivated and assessed within academic programs.

GenAI Literacy and Strategic Use (GC8) refer to understanding GenAI capabilities and limitations and using tools responsibly and efficiently to realise strategic application (S#04; S#29; S#42). Many data sources stressed the importance of recognising hallucinations, bias, and discriminatory patterns (S#68; S#15; S#18). Institutions should provide structured learning opportunities (e.g. training sessions, workshops, and clinics) to help students develop informed, ethical GenAI use. Equipping students with the critical thinking skills necessary to identify and address these limitations is essential for ensuring that they can make informed and constructive decisions of strategically using GenAI. Emerged as a critical competency, Communication and Confidence (GC9) highlights the ability to convey ideas clearly, confidently, and appropriately across a range of interpersonal and professional contexts. To develop this graduate capability, learning programs must incorporate opportunities to evaluate and build core communication skills. These include email etiquette, negotiation strategies, synthesising key points from discussions, structured conversations, cultural competence, and communication in project management contexts (S#71; S#27; S#06). Embedding such elements within assessments ensures students are prepared to articulate their ideas effectively and professionally in diverse settings. For example, crisis-based assessment scenarios were used to evaluate communication under pressure (S#71). Debates and structured discussions were highlighted as effective methods for strengthening verbal reasoning and confidence (S#27). The next graduate competency identified is Academic Writing and Referencing (GC10). It is essential that university graduates develop the ability to produce coherent, well-structured scholarly texts that accurately and ethically cite relevant sources (S#24; S#08; S#44). The rise of GenAI has introduced challenges such as hallucinated references and inaccuracies (S#44), making it critical for students to learn to evaluate and refine their writing rigorously (S#80; S#28). Although institutional requirements for referencing GenAI vary (S#24; S#25), maintaining high academic integrity standards remains essential.

Figure 6 presents the sub-themes that emerged from key theme T5, emphasising the essential graduate competencies that should be explicitly taught, developed, and assessed within academic ICT programs to equip students as future-ready professionals.

Theme 6—Assessment strategies (T6)

The next major theme to emerge from the findings is *Assessment Strategies* (T6). In this study, assessment strategies refer to structured approaches for designing assessments that align learning outcomes with pedagogical intent, uphold academic integrity, and promote authentic, future-focused learning suited to real-world professional contexts. The following strategies appeared consistently across document transcripts, with emphasis on design considerations relevant to ICT.

Excessive reliance on GenAI can hinder the development of essential foundational knowledge and pose challenges as students enter the workforce, especially in fields requiring deep understanding of core concepts like computational thinking, programming logic, and fundamental computing (S#41; S#77; S#55; S#46). To address this, the first recommended strategy is Assess Foundational Knowledge In-Person (AS1). This strategy emphasises assessing foundational ICT content through supervised, in-person formats, enhancing authenticity and ensuring students have genuinely mastered key principles. Findings also recommended the use of low-stakes assessment tasks such as tutorials, lab exercises, or in-class activities, contribute minimally to final grades. These low-stakes assessments play a critical role in reinforcing foundational knowledge while also mitigating assessment-related anxiety (S#01; S#02; S#12). The findings propose Using GenAI to Tutor Low-Stakes Assessments 24/7 (AS2) as a strategy, that leverages GenAI tools to provide continuous, on-demand tutoring support, allowing students to consolidate foundational concepts at their own pace (S#01; S#09; S#75; S#37). Ungrading

Fig. 6 Sub-themes emerged from the Key Theme T5

for Low-Stakes Assessments (AS3) is another assessment strategy which emphasises feedback-driven formative practice, allowing multiple attempts to develop understanding until mastery is achieved (S#51). This formative approach is particularly suited to low-stakes tasks, where traditional one-time grading is replaced with feedback-driven methods. It encourages student reflection, and supports the development of deeper conceptual understanding, learning from the mistakes. Another assessment strategy related to low-stakes assessment is Conduct Low-Stakes Assessments Frequently (AS4). This approach advocates for the regular implementation of low-pressure assessments throughout the semester to monitor student progress, reinforce learning, and alleviate performance-related anxiety (S#01; S#12; S#09). A key design consideration is to provide students with multiple, progressively structured assessment opportunities that support the gradual and sustained development of foundational knowledge.

One of the most frequently cited strategies is Implement Multi-Staged, Connected Scaffolding (AS5), which structures assessments as sequential, interconnected tasks supported by ongoing feedback, ultimately culminating in a final project (S#69; S#29; S#25; S#19; S#14; S#09; S#32). This approach deepens learning and reduces misconduct risks (S#19). Extending AS5, Assisted Assessment Drafting & Planning (AS6) integrates GenAI during early planning stages (e.g., generating ideas or outlines) promoting iterative development while supporting responsible GenAI use (S#29; S#17; S#10; S#16). Assess Process Over Product (AS7) assessment strategy calls for assessing the workflow, not only the final output. This structured approach designs assessments as a sequence of interconnected tasks with guidance at each stage, supporting learning and building competence. Through cycles of submission, feedback, and revision, students' progress toward higher-order objectives. Assignments are sequenced to culminate in a final project (S#32). Students may submit GenAI prompts (S#69), reflective commentaries (S#48; S#29), planning notes (S#29; S#25) or time-stamped drafts (S#15). This mirrors real-world documentation and enhances integrity (S#52; S#33; S#28; S#14).

Aligned with Bloom's Taxonomy, the strategy Assess Higher-Order Thinking Over Theoretical Recall (AS8) advocates for designing assessments that evaluate students' critical thinking, analysis, and problem-solving abilities, rather than tasks focused purely on memorisation or recall of theoretical content (S#75; S#71; S#25; S#66; S#11).

This approach highlights the importance of carefully crafting assessment questions, as surface-level prompts are easily addressed by GenAI and therefore should be avoided in order to maintain academic rigor (S#69; S#25; S#31). Personalise Assessments with Authentic Contexts (AS9) is another widely discussed strategy. It tailors the assessments to real-world scenarios and individual student backgrounds to enhance relevance, engagement, and deeper learning (S#75; S#40; S#31; S#29; S#27; S#23). This includes using disciplinary, situational, and student-specific questions, for example, asking students to reflect on personal experiences, respond to case studies, or connect course content to their own lives (S#32; S#25; S#19).

Another innovative assessment strategy proposed is Facilitate Collaboration with GenAI (AS10). This approach reimagines group work by integrating GenAI tools to support collaborative learning and reflection. It aims to foster intelligent dialogue among team members, promote higher-order thinking, and mirror real-world industry practices involving human–AI collaboration (S#16; S#43; S#06; S#07; S#14). For instance, AI-facilitated virtual study groups could enable students to work jointly on projects, with the AI assigning roles based on individual strengths, monitoring progress, and providing timely hints and feedback (S#58; S#16). Implement Flipped AI-Driven Assessments (AS11) is an instructional strategy where foundational content is learned independently, often through AI-driven pre-assessment activities, allowing in-class time to focus on application, analysis, and higher-order thinking (S#13; S#75; S#08). This approach leverages GenAI tools to support student-led learning in low-stakes assessment tasks, which are then deepened and refined through interaction with the human teacher, leading to more advanced learning outcomes.

Restrict Assessment Open Time (AS12) strategy limits assessment availability windows to reduce misconduct and develop time-management skills, reflecting real-world decision-making pressure (S#21; S#31; S#14; S#10). Facilitate a Continuous Feedback Loop (AS13) strategy emphasises the importance of providing ongoing, interactive feedback between teachers and students to support iterative learning, encourage continuous improvement, and help authenticate student work (S#36; S#19; S#16; S#08). Integrate Multimodal Artefacts Innovatively (AS14) is another prominently discussed assessment strategy. It involves incorporating diverse media formats, such as vlogs, podcasts, graphic abstracts, posters, and short films, alongside traditional written or oral assessments to cater to varied learning styles and showcase a broader range of student skills (S#32; S#31; S#29; S#25; S#19). This triangulated approach not only enhances academic integrity by encouraging authentic student work but also increases student motivation. As these formats are seen as directly relevant to real-world employability, students are less likely to outsource them, and the assessments contribute to the development of more well-rounded competencies (S#25; S#15).

GenAI De-Scaffolding (AS15) strategy gradually reduces GenAI support across tasks to foster independence, for example, early assessments may allow extensive AI use, with later tasks requiring autonomous work (S#32; S#29; S#28). Include Essential Critical Thinking Aspects (AS16) strategy embeds tasks that ask students to question assumptions, consider multiple perspectives, and develop well-reasoned arguments, supporting higher-order cognitive development (S#75; S#58; S#18; S#31; S#20; S#80). Teachers Use GenAI for Assessment Diagnostics (AS17) strategy encourages educators to test assessments with GenAI before deployment, ensuring tasks require thinking beyond what AI can easily produce (S#25; S#14; S#07; S#05; S#03). Finally, Enforce Supervision/Invigilation for Gatekeeping Assessments (AS18) strategy involves conducting milestone assessments, such as capstone projects, critical technical units, and key end-of-semester tasks in controlled, supervised environments (S#26; S#16; S#12). The purpose is to ensure students have genuinely achieved the intended learning outcomes while maintaining academic integrity. These high-stakes, invigilated assessments act as secure checkpoints within the curriculum, fostering confidence that students who successfully complete them have done so independently.

Figure 7 presents the sub-themes that emerged from key theme T6, outlining the assessment strategies identified in the findings as suitable for the GenAI era and recommended for inclusion in academic ICT programs.

Theme 7—Assessment methods (T7)

The final theme that emerged from the data is *Assessment Methods* (T7). In this study, assessment methods refer to the tools, instruments, or techniques used to gather evidence of students' achievement of learning outcomes. These methods provide tangible means for evaluating performance and making informed academic judgments. The findings identified a range of assessment methods suitable for the GenAI era, selected based on their alignment with learning outcomes. These are detailed below.

Process Documentation with Iterative Evidence (AM1) requires students to submit step-by-step records of their work, reflecting their learning progression and decision-making (S#52; S#48; S#33; S#29; S#28; S#25; S#14). This aligns with *Assess Process over Product (AS7)*, valuing reflective practice over final output. Annotated Bibliographies (AM2) involve students listing sources with reflective annotations explaining their relevance and contribution to research or argument (S#48; S#25; S#15; S#13; S#09; S#27). This supports critical evaluation, particularly in research tasks. Debugging or Code Tracing Tasks (AM3) assess students' ability to trace, analyse, and correct code, often through manual simulation (S#71; S#50; S#49; S#46; S#72). Common in programming units, this method promotes hands-on, in-class engagement. Hackathon-Style Challenges (AM4) are in-class assessments used in coding or algorithm-based subjects, fostering collaboration, creativity, and problem-solving in time-constrained settings (S#58; S#71; S#78).

Interactive Hands-on Tutorials and Labs (AM5) assess applied skills via engagement with tools and technologies in practical settings (S#59; S#16; S#09). Effective across ICT

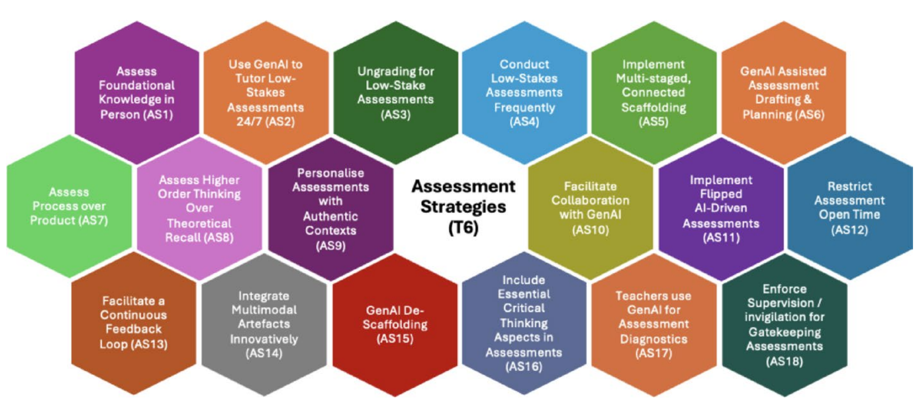

Fig. 7 Sub-themes emerged from the Key Theme T6

domains like networking, programming, and project management. Competency-Based Assessments (AM6), common in vocational education, use a pass/fail rubric to assess mastery of skills, allowing repeated attempts (S#28; S#12). They evaluate practical application of theoretical knowledge (S#43; S#24; S#48). There was strong opposition to traditional exams, which were seen as rigid and ineffective for assessing higher-order skills (S#71; S#27; S#25). As a GenAI-era alternative, *Controlled Exams and Quizzes* (AM7), conducted in supervised, often open-book formats, support academic integrity while fostering conceptual thinking (S#21; S#10; S#71; S#59; S#25). In-Class Concept Mapping, Storytelling, and Flowchart Design (AM8) require students to visualise or narrate ideas in real time, promoting conceptual understanding and communication (S#25; S#02; S#58; S#09; S#32; S#29).

Peer Reviews (AM9) engage students in evaluating each other's work using set criteria, enhancing feedback literacy, critical reflection, and evaluative judgment (S#71; S#31; S#15; S#09). Self-Reflections, Portfolios, and Journals (AM10) involve documenting personal learning journeys, encouraging metacognition and self-assessment (S#09; S#01; S#15; S#43). Portfolios compile student work over time (S#27; S#02; S#25). Experiential Project-Based Learning (AM11) assesses students' application of theory in practical projects, individually or in teams (S#27; S#23; S#15; S#09). These projects often culminate in a functional artifact and report, showcasing technical and soft skills (S#27; S#15; S#67). Authentic or Context-Based Assessments (AM12) are rooted in real-world or simulated scenarios, such as case studies or TED-style presentations, to evaluate applied knowledge and transferable skills (S#34; S#43; S#27; S#01; S#13; S#23; S#12; S#11; S#09). Simulations and Role-Plays (AM13) involve students acting out roles or scenarios (e.g., negotiations or crises) to demonstrate applied competencies and communication under pressure (S#43; S#23; S#78; S#71; S#01; S#06).

Milestone-Based Group Projects (AM14) are structured around staged deliverables, enabling iterative development and feedback over time (S#01; S#07). Interviews, Presentations, and Demonstrations (AM15) assess oral communication, professionalism, and individual contributions through live or recorded formats. Q&A components verify authenticity and depth of understanding (S#33; S#18; S#12; S#31; S#25; S#24; S#63; S#29; S#14; S#10). Real-Life Problem-Solving Assessments (AM16) challenge students to address authentic industry problems using analytical and design thinking, linking classroom learning with real-world application (S#70; S#29; S#15; S#58; S#01). Multimodal Artifacts (AM17) use formats like infographics, podcasts, or social media posts to assess the same outcomes as essays, allowing creative and authentic demonstrations of learning (S#32; S#29; S#25; S#19; S#18; S#09; S#80; S#31). Capstone Projects and Professional Experience (AM18) mark key learning milestones, requiring interdisciplinary knowledge and culminating in a substantial project or industry-based experience (S#16; S#06; S#24; S#02; S#71).

These assess graduate readiness. Multi-Staged Connected Essays or Written Reports (AM19) are modernised essays adapted for the GenAI era. Divided into stages, proposal, draft, peer review, final submission, they foster authentic engagement and progression (S#07; S#27; S#09; S#02; S#17; S#14; S#10). Figure 8 presents the sub-themes under key theme T7, highlighting assessment methods identified as suitable for the GenAI era, based on their alignment with ICT learning outcomes.

Discussions

The discussion is structured around three key sections. The first outlines the Proposed Theoretical Framework, GENESIS (GenAI-Enabled Strategic and Instructional System for Assessment), demonstrating how the seven themes derived from the thematic analysis informed the development of the conceptual model. The second introduces the Assessment Design Matrix (ADM), proposed as an extension of the GENESIS framework to offer a systematic and structured approach for designing targeted, GenAI-enabled assessments. The third section positions the GENESIS Framework and ADM within the scholarly landscape by engaging with relevant literature and comparable models, and it clarifies the rationale for adopting a five-pillar structure over alternative frameworks.

The development of these frameworks was informed by methodological principles suggested by Clarke and Braun (2017), particularly the importance of interpreting each theme's narrative function and its contribution to the overarching research story, as well as the examination of relationships among themes. In addition, the step-by-step thematic analysis procedures recommended by Naeem et al. (2023) for generating conceptual models were applied to identify meaningful interconnections across the themes. Through iterative conceptualisation and interpretation, these thematic relationships converged into a comprehensive framework for transforming assessment practices in ICT higher education.

Proposed theoretical framework

The findings of this study indicate that effective assessment transformation is anchored in seven core themes: *Values and Dispositions* (T1), *Governance* (T2), *Support Infrastructure* (T3), *AI-Integrated Curriculum* (T4), *Graduate Competencies* (T5), *Assessment Strategies* (T6), and *Assessment Methods* (T7). Collectively, these themes illustrate how an educational institution can implement meaningful assessment reform. A key insight

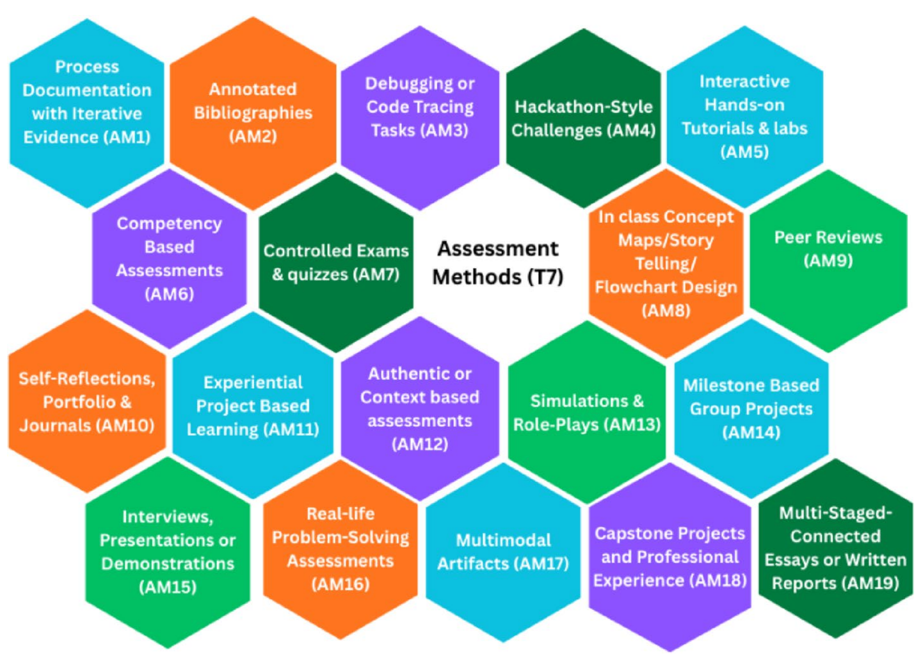

Fig. 8 Sub-themes emerged from the Key Theme T7

is that such reform cannot be confined to academic delivery teams; rather, sustainable transformation requires an institution-wide approach underpinned by aligned values, governance structures, and support infrastructure. Further analysis revealed two overarching patterns that demonstrate how these themes interconnect to guide institution-wide assessment transformation.

The first pattern reflects a sequential developmental pathway across Themes T1 to T4. These themes form a sequential progression, illustrating the foundational conditions required for sustainable transformation. During the write-up, the themes were intentionally re-ordered to reflect this sequence, positioning them as the first four pillars of the proposed GENESIS Framework. This hierarchical pattern shows that assessment transformation must begin with shared institutional values and dispositions (T1), which underpin the development of coherent governance structures (T2). These governance mechanisms then enable the establishment of a supportive infrastructure (T3), upon which curriculum innovation and AI integration (T4) can be effectively advanced. The second pattern emerges across Themes T5, T6, and T7 operate as an integrated, mutually reinforcing set rather than a linear sequence, with each element shaping and supporting the others. T5, T6, and T7 collectively represent the design-level components of assessment practice in the GenAI era. Together, they illustrate how graduate competencies (T5) inform the selection of appropriate assessment strategies (T6), which subsequently shape the design of specific assessment methods (T7). These three themes operate as an integrated matrix, guiding educators in designing assessments. The following paragraphs elaborate the associations among these themes and the rationale for their placement as pillars of the proposed GENESIS Framework.

The thematic analysis indicates that *Values and Dispositions* (T1) form the essential starting point for assessment reform in the GenAI era. Evidence from the sub-themes *Behavioural and Attitude Changes (VD4)*, *Adopting a Facilitative Rather than Policing Approach (VD2)*, and *Cultivating a Culture of Innovation (VD1)* demonstrates that meaningful transformation requires the academic community to embrace mindsets that support openness, responsible GenAI use, creativity, and constructive experimentation. These sub-themes collectively underscore that without a foundational shift in attitudes and institutional culture, subsequent reforms to governance, infrastructure, curriculum, and assessment practices cannot be effectively enacted. Accordingly, T1 serves as the conceptual cornerstone of the framework, establishing the values base upon which all other components of assessment transformation are built. Therefore, T1 forms the first pillar of the GENESIS Framework. Recent literature including studies such as Cabellos et al. (2024), Wang et al. (2025), Shen et al. (2025), Hsiao and Tang (2025) and Danaher (2024) echoes similar conclusions, emphasising the importance of positive perceptions, supportive societal values, and GenAI-aligned cultural norms among students and academics for the acceptance and use of GenAI in academic settings. Therefore, values and dispositions conducive to GenAI integration (T1), form the first pillar of the GENESIS framework. This pillar establishes the foundation for a whole-of-institution mindset transformation, encouraging the academic community to embrace the need for change.

The second pillar of the GENESIS framework, which centres on the findings of theme *Governance* (T2), translates the mindset shift established in Values and Dispositions (T1) by embedding it within formal institutional structures. The value shifts cultivated in T1 foster an innovation-ready, trust-based, and responsibility-oriented culture, providing the essential dispositions for meaningful policy reform that is informed by the findings of T2. Pillar 2 builds directly on this foundation by translating the values of T1 into formalised governance mechanisms. Findings from T2, such as those relating to academic integrity and GenAI policy reform, ethical and equitable digital transformation, and assessment privacy, security, and risk management, demonstrate how institutions must codify the dispositions established in T1 into clear, actionable policies. This progression from T1 to T2 illustrates how an institution moves from shared mindsets to structured, system-level commitments. Through policy clarity, risk management processes, ethical safeguards, and equitable access provisions, the second pillar ensures that the value shifts initiated in T1 are sustained, scaled, and consistently enacted across the institution. Findings from this thematic analysis align closely with recent literature, which highlights the critical need for policy reform in response to GenAI integration. For example, studies by Luo (2024), Christodorescu et al. (2024), Chiu (2024), and Pinho et al. (2025) underscore the importance of aligning academic integrity policies with the realities of GenAI. Luo (2024) and An et al. (2025) further stress the necessity of establishing clear, responsible use guidelines. Similarly, research by Khreibi (2025), An et al. (2025), and Christodorescu et al. (2024) supports the revision of institutional policies to ensure equitable access, provide student support, and address emerging risks such as data privacy and security.

The third key theme to emerge from the data, *Support Infrastructure* (T3), underscores the necessity of a comprehensive, institution-wide support system to effectively implement the governance reforms introduced under Theme T2, thereby forming the third pillar of the GENESIS framework. The policy changes and governance mechanisms established in T2 can only be enacted in practice when supported by an enabling environment that includes robust technological infrastructure, targeted staff capability development, and coordinated academic support structures. This requirement is reinforced by findings from sub-themes such as *Equity and Access for Resources* (SI3), *Upskill Academics* (SI1), and *AI-Enabled Tutoring Support* (SI6), which collectively highlight the institutional conditions needed to operationalise GenAI-aligned governance reforms. In this way, T3 functions as the third pillar of the GENESIS framework, translating governance intent into actionable, operational capacity across the institution. This sequential connection from T2 to T3 illustrates how institutional policies and governance structures provide the framework, but their effective enactment depends on the presence of practical support systems and resources. The findings align with Zlotnikova (2025), who argues that the operational dimension of AI integration requires deliberate investment in infrastructure and professional development. Similarly, Wang et al. (2025), Kangwa (2025), and Rasul (2024) highlight the critical importance of creating adaptive, interactive learning environments that support personalised learning experiences and enable effective GenAI integration in academic practice.

The fourth pillar of the GENESIS framework, which centres on the findings of *AI-Integrated Curriculum* (T4), represents the point at which institutional readiness, established through T1, T2, and T3 is translated into teaching and learning practice. T4 emphasises the integration of GenAI considerations directly into curriculum design, ensuring that academic practices align with the strategic and structural foundations established in the earlier pillars, particularly the need for a robust support infrastructure for both academics and students. The progression from T3 to T4 is therefore both logical and necessary. Findings from sub-themes such as *Quality Gatekeeping Assessments* (AC4), *Programme-Level Assessment Alignment* (AC1), and *GenAI Literacy as a Graduate Attribute* (AC3) reinforce the contribution of T4 to the conceptual framework. Within the GENESIS framework, the fourth pillar represents the translation of institutional readiness into academic practice, ensuring that curriculum design is pedagogically sound, ethically aligned, and technically supported. As the fourth pillar, it marks the point where strategic, structural, and cultural shifts become embedded in the core academic processes, teaching, learning, and assessment, supported by the infrastructure and resources required for effective GenAI integration. Evidence from Zlotnikova (2025), Chen et al. (2025) and Tillmanns et al. (2025) reinforces this interdependence, demonstrating that sustainable GenAI-driven transformation depends on policy alignment, infrastructure readiness, and curriculum redesign functioning as an integrated system.

Findings from the thematic analysis point to the next sequence of themes: *Graduate Competencies* (T5), *Assessment Strategies* (T6), and *Assessment Methods* (T7). These themes extend directly from the *AI-Integrated Curriculum* (T4) and collectively establish the fifth pillar of the GENESIS framework. This progression reflects a clear pedagogical logic. Once GenAI-informed curriculum structures are in place through T4, institutions must ensure that assessment design is aligned with and responsive to the competencies and learning experiences embedded within that curriculum. The data indicate that the curriculum reforms articulated in T4 cannot be sustained without parallel changes to assessment. The relationship among T5, T6, and T7 is not sequential but fundamentally interdependent. The findings of the study indicate that effective assessment design must account for all three themes in an integrated manner. Although *graduate competencies* (T5) provide the foundation that typically guides the selection of *assessment strategies* (T6), which in turn inform the development of specific *assessment methods* (T7), the data show that these elements operate as a mutually reinforcing matrix rather than as a linear progression. Together, they shape assessment design in the GenAI context by ensuring that competencies, strategies, and methods are aligned and responsive to one another. This integrated relationship underpins the Assessment Design Matrix (ADM), an extension of the GENESIS framework that offers a structured and systematic model for developing targeted GenAI-enabled assessments.

Within the GENESIS framework, the integrated relationship among T5, T6, and T7 constitutes the fifth pillar, conceptualised and termed as *Targeted Assessment Design (T5 X T6 X T7)*. This pillar illustrates how assessment practices must be intentionally aligned with curriculum reforms so that the assessments educators design are grounded in GenAI-relevant graduate competencies, supported by coherent assessment strategies, and implemented through methods that are robust, future ready, and responsive to GenAI-driven change. As the culminating stage of the GENESIS framework, Pillar 5 represents the practical enactment of the preceding reforms. It translates institutional values (T1), governance structures (T2), support infrastructure (T3), and GenAI-integrated curriculum design (T4) into assessment practices that are coherent, contemporary, and pedagogically rigorous. The fifth pillar completes the assessment transformation cycle by integrating these three critical components into assessment design and directly addresses the core research question of this study: "What assessment strategies, techniques, and methods are most appropriate for ICT master's students in the context of

Table 2 Thematic linkages and their contribution to the development of the GENESIS framework

Logical connections between themes	Association between Themes/Pillers	Position in the theoretical framework
T1 → T2	Translates foundational values into formal policies and institutional practices	Pillar 1 → Pillar 2 (Values and Dispositions → Governance)
T2 → T3	Converts governance directives into robust support infrastructures that enable consistent implementation	Pillar 2 → Pillar 3 (Governance → Support Infrastructure)
T3 → T4	Provides the infrastructure and resources required for integrating AI effectively into the curriculum	Pillar 3 → Pillar 4 (Support Infrastructure → AI Integrated Curriculum)
T4 → (T5XT6XT7)	Aligns GenAI-integrated curriculum structures with a coherent matrix of competencies, strategies, and methods that collectively shape targeted assessment design	Pillar 4 → Pillar 5 (AI Integrated Curriculum → Targeted Assessment Design)

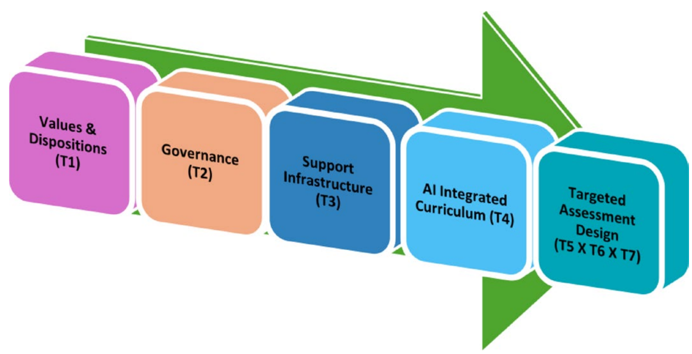

Fig. 9 GENESIS—Five-pillar assessment transformation framework

GenAI?". Table 2 summarises how the themes informed each component of the GENESIS framework and the ADM, highlighting the logical links that guided their development. Figure 9 presents a visual representation of the GENESIS five-pillar assessment transformation framework, illustrating the proposed sequential relationships among the themes.

Proposed assessment design matrix (ADM)

The ADM provides practical guidance for redesigning assessments in an AI-integrated curriculum. It is an extension of the fifth pillar (*Targeted Assessment Design*) of the GENESIS framework and draws on insights from themes T5, T6, and T7. The ADM operationalizes the connection between graduate competencies, assessment strategies, and assessment methods, offering a structured, tripartite approach to assessment design and implementation.

The thematic analysis indicates that the newly identified *Graduate Competencies* (T5), the core capabilities students are expected to demonstrate in an AI-integrated curriculum, must not only be clearly articulated and embedded within the curriculum but also systematically assessed to ensure genuine attainment (S#54; S#48; S#66; S#59; S#57; S#77). The findings further highlight that once the curriculum has been reformed to integrate AI considerations (T4), assessment strategies must be carefully designed and assessment methods thoughtfully implemented so that they align with, and validly measure, the targeted graduate competencies addressing the learning outcomes arriving from the curriculum. For example, competencies such as *Prompt Engineering* (GC1), *Metacognitive Skills* (GC2), and *GenAI Literacy and Strategic Use* (GC8) require the deliberate selection of appropriate assessment strategies, followed by the implementation of assessment methods that explicitly evaluate these capabilities. To operationalise this alignment, the ADM framework introduces a tripartite structure (T5 × T6 × T7), for assessment reform, comprising design-level alignment, design-to-implementation bridge, and implementation-level alignment. The following sections explain how each level is applied in practice and how they align with the objectives of the GENESIS framework.

Findings from the T6 theme show that the emergence of GenAI necessitates revising traditional assessment strategies to address new challenges such as maintaining academic integrity, preserving core disciplinary knowledge, and providing effective scaffolding. Consequently, new assessment strategies (T6) are required to authentically measure the revised graduate competencies (T5), highlighting a strong reciprocal relationship between T5 and T6 (T5 ↔ T6). Designers should first determine the graduate competencies to be developed and then select strategies capable of validly measuring them. This relationship advocates for a design-level alignment in which graduate competencies (T5) drive the selection of appropriate assessment strategies (T6), and each selected strategy must be checked against the targeted competencies to ensure full alignment. The findings further suggest that this relationship is inherently reciprocal: the assessment strategies chosen for a given task must be reviewed and tested against the competencies they intend to measure to confirm proper alignment. Through this two-way process, supported by pedagogical expertise and creativity, assessment designers can ensure that assessment strategies and graduate competencies are coherently matched in the GenAI era.

Findings then suggest a reciprocal relationship between T6 and T7 (T6 ↔ T7). While T6 focuses on the design of assessment strategies that align with revised graduate competencies, T7 provides the enabling assessment methods, tools, and institutional supports required to implement those strategies effectively. Assessment methods (T7) must be capable of operationalising the intended assessment strategies (T6) by offering practical mechanisms for capturing the types of performance, reasoning, or artefacts that each strategy requires. Assessment strategies (T6) determine the choice and design of assessment methods (T7), and the feasibility and characteristics of these methods, in turn, influence how strategies are enacted in practice. This relationship creates a design-to-implementation bridge. Conversely, the chosen assessment strategy also informs which assessment methods are appropriate, feasible, and valid to implement assessment strategies. This interdependence ensures that assessment strategies are not only pedagogically sound but also implementable, reinforcing a coherent and sustainable approach to assessment design in the GenAI era.

Findings of the study also indicate a reciprocal relationship between T5 and T7 (T5 ↔ T7). Insights from the sub-themes within T7 show that assessment methods, whether tools, instruments, or techniques must generate valid and reliable evidence of students' attainment of the targeted graduate competencies (T5). In turn, the selection and design of these methods should be shaped by the nature and level of the competencies they are intended to measure. This association suggests an implementation-level alignment: assessment methods (T7) must yield valid evidence of students' attainment of the targeted competencies (T5), while the characteristics of those competencies determine which methods are most appropriate and effective. Thus, assessment methods function as the practical means of operationalising graduate competencies, ensuring that competencies are not only articulated at the curriculum level but also made measurable, observable, and actionable through appropriate assessment approaches. Conversely, the nature of the competencies guides the selection of suitable methods, reinforcing the iterative and mutually informing relationship between these two themes.

To clarify how the three components of the ADM interact in practice, Table 3 maps the logical connections between the three themes that constitute the tripartite structure. It outlines the reciprocal relationships identified through the thematic analysis and provides practical guidance on how these alignments should be enacted during assessment design and implementation. Figure 10 presents the ADM framework, which operated as a matrix to guide the targeted assessment design.

The flowchart in Fig. 11 illustrates the practical steps an assessment designer should follow when applying the ADM matrix. The process begins at the design level, where the designer selects one or more graduate competencies to be developed through the assessment. These competencies should be drawn from the ten sub themes (GC1 to GC10) listed in Table 1. The next step is to choose assessment strategies that can validly measure those competencies, selecting from the eighteen sub themes (AS1 to AS18) in Table 1. As noted earlier, this stage may require iterative refinement to ensure strong alignment. At the design to implementation bridge, the selected assessment strategies must then be translated into appropriate assessment methods. Additional checks are required at this point to confirm operational compatibility and feasibility, specifically whether the chosen methods can produce valid evidence of students' attainment of the targeted competencies. Suitable methods should be selected from the nineteen options (AM1 to AM19) listed in Table 1. At the implementation level, the chosen assessment

Table 3 Logical connections of the tripartite structure and practical implementation guide

Logical connections of the tripartite structure	Reciprocal associations between Themes/Pillers	Practical implementation guide of the tripartite structure
T5 ↔ T6 (Design-level alignment)	Graduate competencies (T5) should guide the selection of assessment strategies (T6), and each strategy should be checked to ensure that it is aligned with and provides valid measurement of the intended competencies	When designing an assessment, academics must first identify the graduate competencies that are targeted in that assessment and then select assessment strategies that can validly measure those competencies. Conversely, when an assessment strategy is chosen, it must be checked against the targeted competencies to ensure alignment
T6—T7 (Design-to-implementation bridge)	Assessment strategies (T6) determine the choice and design of assessment methods (T7) to operationalise them, and the feasibility and characteristics of the methods, in turn, influence how strategies are implemented effectively	When planning an assessment, academics must translate the selected assessment strategies into appropriate assessment methods that can operationalise those strategies in practice. Conversely, each chosen method should be checked to ensure it appropriately reflects and enacts the intended strategy, maintaining coherence between design and implementation
T5 ↔ T7 (Implementation-level alignment)	Graduate competencies (T5) guide the selection and use of assessment methods (T7) to ensure valid evidence of learning, and the nature of the competencies, in turn, determines which methods are most appropriate and effective	When implementing an assessment, academics must ensure that the chosen assessment methods can generate valid evidence of students' attainment of the targeted graduate competencies, and conversely, each method should be reviewed against the nature and level of those competencies to confirm that it is appropriate and effective for measuring them

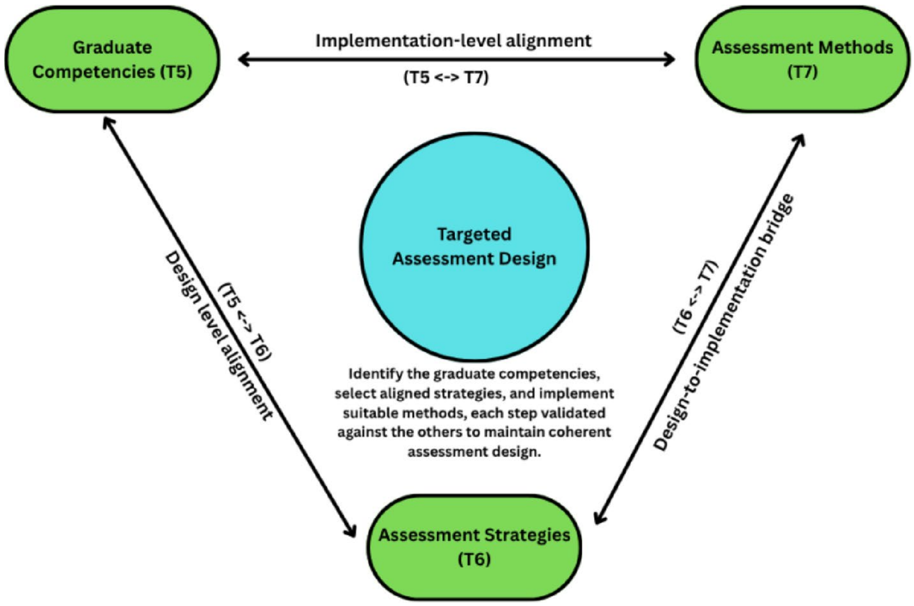

Fig. 10 Assessment design matrix (ADM) showcasing the tripartite structure to guiding the assessment design

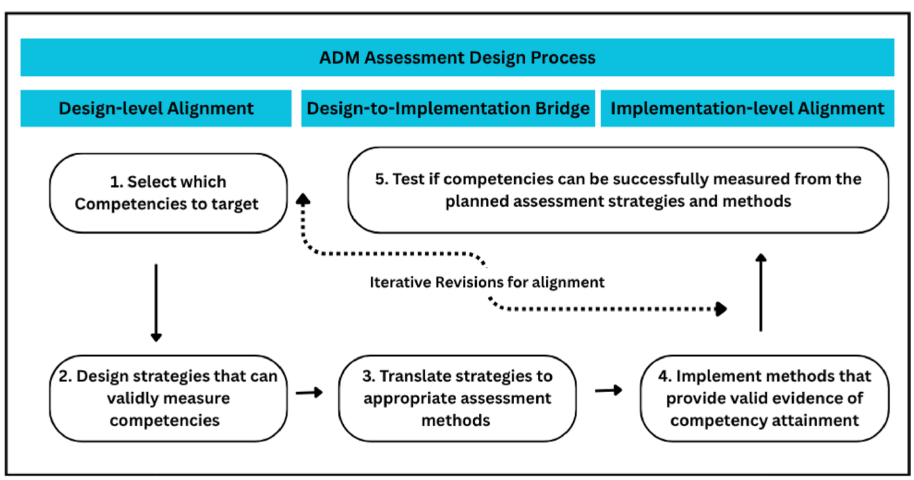

Fig. 11 Assessment Design Matrix (ADM)—Assessment Design Process

methods are applied, with final verification that they generate valid evidence of the intended competencies. Overall, this tripartite structure, graduate competencies, assessment strategies, and assessment methods, supports reciprocal relationships. Assessment designs may therefore undergo several iterations until optimal alignment is achieved.

The core requirement of the ADM is that at least one element from each of its three dimensions must be selected, ensuring that every assessment incorporates all essential components and remains aligned with GenAI-integrated curriculum objectives. As a guiding framework, the ADM acts as a design compass, supporting educators in creating assessments that respond effectively to GenAI's affordances. The following example illustrates how an assessment designer can apply the ADM to develop a sample assessment.

This example is drawn from an IT project management unit in which the primary objective is to design an assessment that addresses the project management learning outcomes while giving particular attention to the graduate competency of *Teamwork* *and conflict resolution* (GC6). In this case, the designer determines that an IT project management mini project sprint provides a suitable assessment strategy for evaluating how students engage in teamwork and manage conflict throughout the project. The designer recognises that a single point of evaluation would not capture the complexity of these behaviours and therefore determines that the assessment strategy must focus on the strategy *Assess process over product* (AS7). When progressing from design to implementation, the designer selects assessment methods capable of generating valid evidence of student performance over time. The chosen approach draws on the requirement for continuous *process documentation supported by iterative evidence* method (AM1). Within this assessment, the designer adopts a method consistent with AM1 in which students produce a process portfolio that documents their team charter and defined roles, their recorded conflicts and resolutions, their decision-making records, their individual reflections, and their summaries of meetings and communication. This approach provides a rich evidence base that captures the evolution of teamwork and conflict resolution throughout the project and is therefore well suited to assessing the targeted graduate competency GC6 within the ADM framework. To verify the designed assessment, the alignment and mapping should be reviewed to ensure that the process documentation adequately evaluates the steps students undertake. The final assessment design should also be checked to confirm that the chosen strategy and assessment method provide sufficient evidence for determining whether students have achieved GC6. Figure 12 presents a visual diagram illustrating how the tripartite structure was used to design an example assessment.

Table 4 presents six illustrative assessment briefs for ICT units, each developed by combining different ADM dimensions. These examples show how the ADM operates in practice and reflect ICT needs, as the thematic analysis and the competencies in Table 1 were grounded in ICT assessment contexts.

Justification of the GENESIS framework and ADM

Only a limited number of existing frameworks engage substantively with components of the intersection between GenAI, institution-wide transformation, and assessment design. However, none provide coverage of the comprehensive scope articulated in the GENESIS and ADM frameworks. Instead, the frameworks reviewed below address isolated dimensions of this intersection or offer only partial alignment with its broader intent.

Nguyen et al. (2025) proposed a framework for AI integration that emphasises social–emotional development, interaction and dialogue, feedback and support, technical accessibility, digital resources, and equity. These factors reinforce the findings of the present study and demonstrate conceptual alignment with several pillars of the GENESIS framework, including values, governance structures, institutional support systems, and curriculum reform. However, GENESIS extends beyond these dimensions by introducing a sequential developmental progression across its pillars, establishing the foundation for a tripartite structure at the assessment design level. This integrated, multi-layered architecture distinguishes GENESIS as a more systemic and scalable framework for AI-enabled curriculum and assessment transformation. Another recent framework relevant to generative AI integration in assessment is the AI-Driven Assessment Framework (Three-Branch Model) by Ilieva et al. (2025). While both the Three-Branch Model and

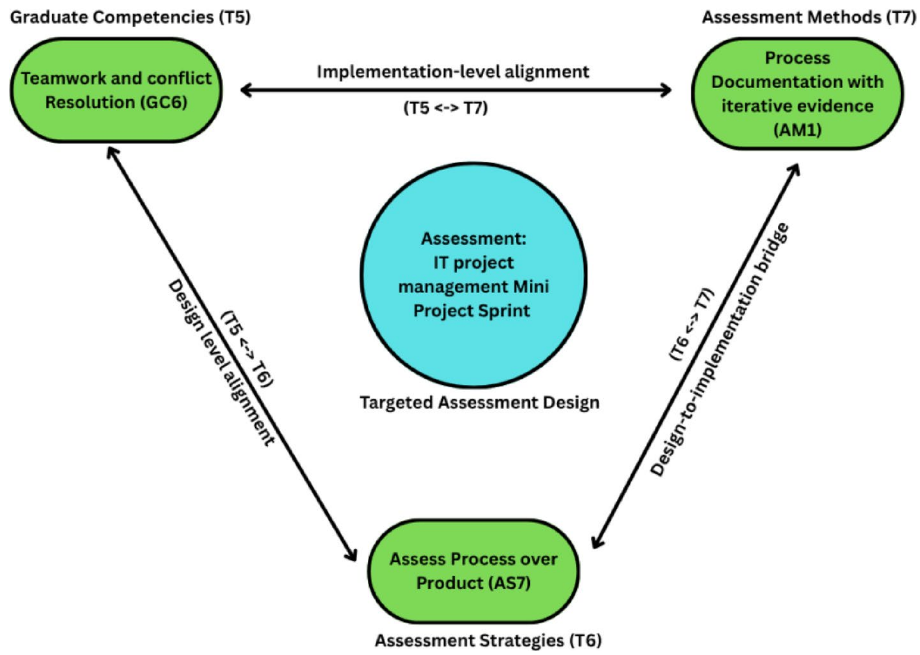

Fig. 12 A practical example of designing an assessment using Assessment Design Matrix (ADM)

GENESIS address GenAI-enabled assessment, they operate at different levels of scope and transformation. The Three-Branch Model centres on the stakeholder ecosystem, specifically instructors, students, and institutional control bodies, and focuses on how these actors engage with AI-enabled assessment practices. In contrast, GENESIS extends far beyond task-level or role-specific considerations, offering a systematic blueprint for institutional transformation across seven interconnected themes spanning values, governance, support structures, curriculum, and assessment design. Whereas the Three-Branch Model emphasises stakeholder responsibilities, GENESIS foregrounds deeper organisational change. The GAISE framework (Baskara, 2024) similarly contributes to GenAI scholarship by outlining how AI can be integrated into curriculum design, teaching methodologies, assessment strategies, and teacher professional development. Its four components (AI-Enhanced Content Generation, Adaptive Personalization, Interactive Knowledge Construction, and Reflective Transformation) emphasise pedagogical innovation and learner-centred AI-enabled instruction. In contrast, the GENESIS framework adopts a broader, institution-wide perspective, positioning assessment reform within a five-pillar organisational structure. While GAISE focuses on how AI transforms teaching and learning within classroom contexts, GENESIS addresses the systemic conditions required for governance-oriented, organisation-level assessment transformation.

Table 4 Sample assessment briefs developed using Assessment Design Matrix (ADM)

Proposed Assessment	Graduate Competency	Assessment Strategy	Assessment Method	Proposed GenAI Aligned Assessment Design Details
IT project management Mini Project Sprint	Teamwork and conflict Resolution (GC6)	Assess Process over Product (AS7)	Process Documentation with iterative evidence (AM1)	Students will engage in a collaborative, time-bound project sprint where they must manage a simulated project lifecycle in teams. The emphasis is on documenting how they worked together. Each team will plan and execute a mini-project over two weeks Students will submit a "Process Portfolio" including Team Charter & Roles, Conflict Logs & Resolutions, Decision-Making Records, Reflections from each member, Meeting Summaries & Communication Evidence
AI-Enhanced Programming Tutorials for Python Programming	Prompt Engineering (GC1)	Use GenAI to Tutor Low-Stakes Assessments 24/7 (AS2)	Interactive Hands-on Tutorials & labs (AM5)	Students will engage in interactive, hands-on lab sessions where they will practice crafting effective prompts to solve programming challenges using AI tools. Students will complete guided coding exercises supported by GenAI. Students will experiment with prompts to refine AI-generated responses, reflect on prompt quality, AI accuracy, and their learning process, receive immediate feedback from GenAI and in-class facilitation. It fosters both technical fluency and ethical awareness in using AI tools responsibly Following deliverables shall be produced: Weekly prompt logs and annotated code outputs, Reflection snippets on prompt effectiveness and In-class participation in live coding labs
Personalised Data Science project	Ethical Intelligence (GC7), GenAI Literacy & Strategic Use (GC8)	GenAI De-Scaffolding (AS15), Personalise Assessments with Authentic Contexts (AS9)	Experiential Project Based Learning (AM11), Interviews, Presentations or Demonstrations (AM15)	Students will work independently on a personalised data project grounded in an authentic context of their choice, healthcare, climate, business, or social justice, reflecting their interests or career goals The project is structured using a GenAI De-Scaffolding approach. Initial stages provide curated guidance and exemplar uses of GenAI, Support is gradually withdrawn as students develop autonomy and critical judgment. Students are expected to make strategic, ethical decisions about when and how to use GenAI for tasks like data cleaning, hypothesis generation, analysis, or communication. Students will experience the full data science lifecycle from problem definition and data wrangling to model evaluation and ethical reflection. They will document their process and reflect on the ethical, strategic, and technical choices made, especially regarding GenAI use. Deliverables include Proposal in early stage, final Project Report, Interactive dashboard or final data product and 5-min demonstration

Table 4 (continued)

Proposed Assessment	Graduate Competency	Assessment Strategy	Assessment Method	Proposed GenAI Aligned Assessment Design Details
Business Communication Assignment – simulated workplace conflict scenario	Communication & Confidence (GC9), Emotional intelligence (GC4)	Assess Foundational Knowledge in Person (AS1)	Simulations & Role-Plays (AM13)	Students will be assessed on their communication & confidence and emotional intelligence as they respond in real time to emotional cues, de-escalate conflict, and maintain professionalism under pressure. The task is delivered through Simulations & Role-Plays, where each student takes the role of a customer service representative in a IT service delivery organisation in a mock scenario with a tutor or peer acting as an client. Scenarios are designed to mimic real industry complaints, delays, billing errors, or service dissatisfaction, requiring adaptive, empathic communication strategies. Live role-play performance, reflective report piece and peer review or rubric-based feedback
Cloud Computing Adoption in SMEs: Barriers, Benefits, and Strategic Implications—Essay	Academic Writing and Referencing (GC10), Metacognitive Skills (GC2)	Implement Multi-staged, Connected Scaffolding (AS5), GenAI Assisted Assessment Drafting & Planning (AS6)	Multi-Staged-Connected Essays or Written Reports (AM19), Annotated Bibliographies (AM2)	This assessment invites students to explore a key topic in cloud computing through a multi-staged, connected essay process designed to build both their academic Writing & Referencing and Metacognitive Skills. Using GenAI-assisted planning tools, students will iteratively develop an academic essay with embedded self-reflection checkpoints. The process will be supported by scaffolded stages including topic proposal, annotated bibliography, AI-assisted draft or outline with used prompts, peer feedback, and final submission, encouraging students to reflect on how they think and learn throughout the writing journey
Data structures and Algorithems – Mid semester test	Decision-making (GC5)	Assess Foundational Knowledge in Person (AS1), and Enforce Supervision / invigilation for Gatekeeping Assessments (AS18)	Controlled Exams & quizzes (AM7)	This controlled, in-person mid-semester test is designed to assess students' foundational understanding of core data structures and algorithmic techniques through timed higher order tasks. Aligning with the competency of decision-making students will be required to select and justify the most efficient solutions under pressure, simulating real-world problem-solving scenarios. Delivered via Controlled Exams & Quizzes this assessment supports the Foundational Knowledge Assessment Strategy by validating students' core logic, structure selection, and algorithmic reasoning skills in a monitored setting

The findings of this study especially from the ADM framework demonstrate strong theoretical alignment with established assessment design frameworks, notably Constructive Alignment (CA) (Biggs, 1996) and Evidence-Centered Design (ECD) (Mislevy & Haertel, 2006). ECD follows a structured process, from defining competencies to designing assessment methods and interpreting results. Within ECD, Graduate Competencies (T5) align with Domain Analysis and Modeling, Assessment Strategies (T6) correspond to the Conceptual Assessment Framework, and Assessment Methods (T7) map to Assessment Implementation (Mislevy & Haertel, 2006). Similarly, in the Constructive Alignment (CA) model, Graduate Competencies (T5) correspond to Intended Learning Outcomes, Assessment Strategies (T6) align with Teaching and Assessment Activities, and Assessment Methods (T7) relate to Assessment Tasks. The proposed ADM framework is theoretically aligned with the tripartite dimensions of both CA and ECD, thereby demonstrating its validity alongside these established models and reinforcing the rigor of the data analysis. ADM aligns with CA's outcome-driven design by connecting learning outcomes with instructional activities and assessments, promoting coherent curriculum planning, and emphasising constructive alignment. Similarly, ADM reflects ECD's focus on capturing knowledge, skills, and abilities (KSAs), the evidence of these KSAs, and tasks that elicit valid evidence, thereby grounding assessment in observable proof of competency. The core alignment among competencies, strategies, and assessment methods is justified, and ADM further extends these foundational frameworks to contemporary, technology-enhanced, and employment-focused learning environments.

While ADM is grounded in the theoretical foundations of CA and ECD, it distinguishes itself by operationalising these principles at the level of course assessment design, with a stronger emphasis on practical implementation. CA and ECD function as generic theoretical frameworks, whereas ADM explicitly links learning design to graduate competencies and industry-relevant skills, particularly within the ICT discipline. While ECD primarily emphasises formative assessments, ADM integrates both formative and summative strategies. Unlike CA and ECD, which remain largely theoretical, ADM provides step-by-step guidelines for designing and adapting assessments to address challenges such as maintaining academic integrity and accurately evaluating student learning in the GenAI era. ADM also incorporates a verification process to ensure sufficient evidence of competency attainment and is context-responsive, addressing disciplinary needs, and the affordances of GenAI. By offering a flexible scaffold for eliciting and validating learning evidence, ADM enables assessment designers to combine strategies and methods creatively, particularly in complex or rapidly evolving ICT contexts. In doing so, it synthesises the strengths of CA and ECD into a practical, technology-aware framework that bridges the gap between theoretical rigor and contemporary, competency-focused assessment practice.

Shanto et al. (2025) proposed the ACE framework, which organises the integration of GenAI models to facilitate advanced analysis, synthesis, and evaluation, as outlined in Bloom's taxonomy, progressing from basic knowledge recall to complex conceptualisation. Both ACE and ADM operate on the shared principle that assessments should target higher levels of learning. While the steps outlined in the ACE framework theoretically align with elements of ADM, such as defining and mapping learning outcomes to assessment activities, ADM provides a more practical and systematic guide for designing and validating assessments within an AI-integrated curriculum. Through its tripartite structure, ADM offers clear mechanisms to ensure that AI-era competencies are explicitly targeted, validly measured, and coherently enacted in practice. ADM therefore extends beyond the cognitive focus of ACE, delivering a comprehensive and actionable framework for assessment reform in the GenAI era.

Conclusions, implications and limitations

This study aimed to determine how educators can design assessments for ICT tertiary students that accurately measure learning outcomes in an era of widespread access to GenAI tools. In pursuing this aim, the research examined the assessment design and implementation challenges that GenAI introduces to discipline-specific assessments in higher education from an academic institutional perspective. By thematically analysing 80 publicly available web-based academic and institutional documents, including existing assessment models, frameworks, pedagogical practices, and guidelines, the paper directly addressed the central research question. The thematic analysis that has been undertaken followed Braun and Clarke's (2006) six-phase thematic analysis framework. The thematic analysis yielded 7 key themes and 70 sub themes that collectively addressed the research objectives. These insights informed the development of GENESIS and the Assessment Design Matrix (ADM), two complementary tools created to support the transformation of assessment practices and to ensure integrity, validity, and relevance in GenAI-enabled ICT education.

The study's main findings are captured in two complementary models, GENESIS and the Assessment Design Matrix (ADM). GENESIS is a five-pillar framework that outlines how institutions can achieve sustainable assessment transformation in the GenAI era. Its first four pillars (Values and Dispositions, Governance, Support Infrastructure, and AI-Integrated Curriculum) form a sequential progression that builds the cultural, structural, and pedagogical foundations needed for change. The fifth pillar, Targeted Assessment Design, introduces a matrix that links Graduate Competencies, Assessment Strategies, and Assessment Methods, ensuring assessment practices are coherent, aligned, and responsive to GenAI-driven shifts. Extending this pillar, the ADM provides a practical, three-level alignment model for designing and implementing assessments within an AI-integrated curriculum. It clarifies how competencies inform strategy selection, how strategies guide method choice, and how methods generate valid evidence of competency attainment. Together, GENESIS and ADM illustrate how institutional conditions and assessment-level decisions interact to produce valid, ethical, and future-ready assessment practices in ICT higher education.

This study contributes to the field by offering a discipline-specific, practical guide for transforming assessment in ICT education at a time of rapid GenAI adoption. It highlights that meaningful reform requires an institution-wide approach involving shifts in values, governance, mindsets, and support systems. The GENESIS and ADM frameworks provide educators with actionable tools that connect graduate competencies to suitable strategies and methods, producing an industry-aligned and pedagogically robust approach to assessment design. Furthermore, the two frameworks complement Australia's TEQSA National Implementation Toolkit (organised under the domains of process, people, and practice) by translating its high-level principles across governance, academic integrity, risk mitigation, and staff capability into concrete, practice-oriented processes. By operationalising these national expectations through detailed guidance for assessment design, GENESIS and ADM offer a cohesive pathway for aligning institutional practice with sector standards while meeting the evolving needs of ICT education in the GenAI era.

Implementing the GENESIS framework and ADM, however, presents practical challenges, including staff resistance, resource limitations, workload pressures, and ethical considerations related to GenAI in assessment. Targeted faculty development, supportive policies, and institutional readiness strategies are therefore essential. Central to GENESIS is a whole-institution approach that begins with Values and Dispositions (T1), the framework's conceptual foundation. This pillar creates the conditions for meaningful transformation by promoting openness, responsible GenAI use, creativity, and a culture of innovation, directly addressing issues such as resistance to change. Without this foundational shift in attitudes and institutional culture, reforms to governance, infrastructure, curriculum, and assessment practices cannot be effectively sustained. By cultivating a culture aligned with the framework's values and reinforcing it through supportive structures, institutions can strengthen their capacity for adoption, ensuring that assessment transformation is both feasible and sustainable.

The ADM was developed with a focus on ICT assessment design; however, its principles are readily transferable to other disciplines. Once designers identify the relevant graduate competencies for a particular field or a context, suitable assessment strategies and methods can be selected, enabling both the GENESIS framework and the ADM to extend beyond ICT. Designed to be scalable and adaptable, the ADM can be customised to program-specific requirements and updated as curriculum or industry needs evolve, simply by revising the graduate attributes associated with a given degree. This flexibility supports continuous assessment transformation and the creation of innovative, future-ready assessments across diverse educational contexts, while also advancing theoretical understanding of GenAI's role beyond a technical tool toward a more holistic model of institutional integration.

Despite its contributions, this study has some limitations. The derived lists of competencies, strategies, and methods are constrained by the scope of the analysed documents, potentially omitting elements not captured in the thematic analysis. Data collection was limited to qualitative, web-based sources published within a short timeframe, and publicly accessible policies and reports may not reflect internal practices, tacit knowledge, or contextual nuances within institutions. As a result, the GENESIS framework and ADM represent formalised, outward facing elements of practice that may have limited applicability in settings where internal processes differ. Future research should therefore empirically validate both models through pilot implementations, structured feedback from educators and students, and the inclusion of internal documents and practitioner insights to enhance contextual relevance. Longer term validation using longitudinal and mixed methods approaches will be important for assessing their effectiveness across diverse institutional and disciplinary contexts. Further work is also needed to examine effective faculty development strategies and to explore how the framework and ADM can be adapted to the needs of different academic contexts, strengthening their robustness, adaptability, and contribution to sustainable assessment reform in the era of GenAI.

Appendix 1

Index	Document Title	Document Type	Source / Author
1	Rethinking assessment strategies in the age of artificial intelligence (AI) (Assessed: 5 Mar 2025)	Institutional Policy/procedure	Charles Sturt University
2	Assessment and Generative Artificial Intelligence (Accessed: 5 Mar 2025)	Institutional Policy/procedure	Charles Sturt University
3	Assessing Generative Artificial Intelligence Resilience in assessments (Accessed: 5 Mar 2025)	Institutional Policy/procedure	Charles Sturt University
4	Making informed decisions about Generative Artificial Intelligence (GenAI) use in teaching and assessment (Accessed: 5 Mar 2025)	Institutional Policy/procedure	Charles Sturt University
5	Embracing Generative AI in Assessments: A Guided Approach (Published: 14 Aug 2024)	Guidance Document	JISC / Sue Attewell
6	Aligning our assessments to the age of generative AI (Published: 26 Nov 2024)	Institutional Policy/procedure	University of Sydney / Adam Bridgeman, Ruth Weeks and Danny Liu
7	AI and assessment (Accessed: 5 Mar 2025)	Institutional Policy/procedure	Monash University
8	Making sense of generative AI for assessments: Contrasting student claims and assessor evaluations (Published: November 2023)	Journal Article	Isabel Fischer, Simon Sweeney, Matthew Lucas, Neha Gupta
9	Rethinking assessment in response to AI (Published: July 2023)	Institutional Policy/procedure	University of Melbourne/ Melbourne Centre for the Study of Higher Education
10	AI and assessment: Rethinking assessment strategies and supporting students in appropriate use of AI (Published: 13 May 2024)	Guidance Document	My chartered College / Lynsey Meakin
11	Race with the machines: Assessing the capability of generative AI in solving authentic assessments (Published: 2023)	Journal Article	Thanh, B. H. et al
12	Assessment Considerations for Course and Program Development (Published: Aug 2024)	Institutional Policy/procedure	UNSW Sydney
13	Checklist for self-auditing assessments in the world of AI (Published: 12 Oct 2023)	Institutional Policy/procedure	UNSW Sydney
14	Assessment Control Factors AND Categories of permitted AI use in assessment (Accessed: 6 Mar 2025)	Institutional Policy/procedure	UNSW Sydney
15	Assessment Strategies (Accessed: 6 Mar 2025)	Institutional Policy/procedure	Johns Hopkins University
16	Assessment reform for the age of artificial intelligence (Published: 23 November 2023)	Guidance Document	TEQSA, Tertiary Education Quality and Standards Agency—Australia
17	Using AI to enhance assessment (Accessed: 6 Mar 2025)	Guidance Document	University of Melbourne/ Melbourne Centre for the Study of Higher Education
18	A scoping review on how generative artificial intelligence transforms assessment in higher education (Published: 24 May 2024)	Journal Article	Xia, Q et al
19	Generative AI tools and assessment: Guidelines of the world's top-ranking universities (Published: 15 December 2023)	Journal Article	Benjamin Luke Moorhouse, Marie Alina Yeo, and Yuwei Wan
20	Assessment and learning outcomes for generative AI in higher education: A scoping review on current research status and trends (Published: 18 Aug 2024)	Journal Article	Weng, X., XIA, Q., Gu, M., Rajaram, K., & Chiu, T. K
21	Strategies for e-Assessments in the Era of Generative Artificial Intelligence (Published: 20 Feb 2025)	Journal Article	Tapiwa Gundu

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23	Contextual assessment design in the age of generative AI (Published: 28 Feb 2025)	Journal Article	Gonsalves, C
24	Reconsidering assessment for the ChatGPT era: QAA advice on developing sustainable assessment strategies (Published: July 2023)	Institutional Policy/procedure	The Quality Assurance Agency for Higher Education, UK
25	Approaches to assessment in the age of AI (Accessed: 6 Mar 2025)	Institutional Policy/procedure	King's College London
26	Why we need to redesign assessments with generative AI (Accessed: 6 Mar 2025)	Institutional Policy/procedure	AUT, New Zealand
27	Rethinking Assessment in Light of Generative AI (Published: 10 Jan 2024)	Blog/Magazine/Social media post	C2C Digital Magazine, David Swisher & Annie El
28	Assessment and Generative artificial Intelligence in higher Education (Published: April 2024)	Research Report	University of Illinois Chicago/Hanover Research
29	Assessment Design using Generative AI (Accessed: 7 Mar 2025)	Guidance Document	The University of British Colombia
30	Revolutionizing Educational Assessments: How Generative AI is Shaping the Future (Published: 24 Feb 2024)	Blog/Magazine/Social media post	Vaikunthan Rajaratnam
31	How to tweak assessments to limit the use of generative AI (Accessed: 7 Mar 2025)	Institutional Policy/procedure	Glion Institute of Higher Education
32	Designing assessments to mitigate the use of AI writing tools (Accessed: 7 Mar 2025)	Institutional Policy/procedure	Brock University
33	Assessment and Feedback in the Generative AI Era: Transformative Opportunities, Novel Assessment Strategies and Policies in Higher Education (Published: Dec 2023)	Conference Proceedings	Souppez, J. B.R.G.1, Goswami, D., and Yuen, J
34	How should we change teaching and assessment in response to increasingly powerful generative Artificial Intelligence? Outcomes of the ChatGPT teacher survey (Published: 26 January 2024)	Journal Article	Matt Bower, Jodie Torrington, Jennifer W. M. Lai, Peter Petocz & Mark Alfano
35	AI as an opportunity to rethink assessment strategy (Published: 26 Feb 2024)	Opinion paper	Murray MacKay, Imperial College London
36	Strategies for Effective Feedback in the Age of Generative Artificial Intelligence (Published: 8 Oct 2024)	Guidance Document	Patrick Kelly, Lorelei Anselmo, & Lin Yu/ University of Calgary
37	The AI Assessment Scale: Version 1 (Accessed: 7 Mar 2025)	Blog/Magazine/Social media post	Leon Furze
38	Student Forums 2023: Exploring the role of generative AI in assessments: A student perspective (Published: 14 June 2023)	Blog/Magazine/Social media post	JISC / Sue Attewell
39	Strategic Approaches for GenAI Adoption in Schools (Published: 11 Mar 2024)	Blog/Magazine/Social media post	Abdulkadir Özbek
40	A Practical Guide to Using Generative AI for Assessment for Learning (Published: 1 Mar 2025)	Guidance Document	Niall McNulty
41	The Negative Impact of AI on Academic Integrity in Tertiary Education (Published: 4 Dec 2024)	Blog/Magazine/Social media post	Dr Michael Coole, Dr Warren Doudle, Nicola Lockhart, and Simeng Yuan

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42	Gen AI strategies for Australian higher education: Emerging practice (Published: Nov 2024)	Guidance Document	TEQSA, Tertiary Education Quality and Standards Agency—Australia
43	Redesigning Assessments for AI-Enhanced Learning: A Framework for Educators in the Generative AI Era (Published: 2 Feb 2025)	Journal Article	Zuheir N. Khlaif 1,Wejdan Awadallah Alkouk, Nisreen Salama and Belal Abu Eideh
44	Exploring the integration of Generative AI in assessment in a tertiary context: a Case study (Published: Dec 2024)	Conference Proceedings	Eleni PETRAKI
45	Understanding Student and Academic Staff Perceptions of AI Use in Assessment and Feedback (Published: 22 Jun 2024)	Journal Article	Jasper Roe, Mike Perkins, and Daniel Ruelle
46—CS	Assessment in Computer Science Education in the GenAI Era—Prioritizing critical thinking over syntax mastery (Published: 10 Jan 2025)	Blog/Magazine/Social media post	Orit Hazzan, Communications of the ACM
47	Generative AI vs Assessments (Accessed: 8 Mar 2025)	Guidance Document	Scott Daniel, and Ghulam Mubashar Hassan, ACDICT
48	Case Study: Generative AI adapted assignments in Computer Science (Published: 2 Sep 2024)	Guidance Document	Professor James Davenport, University of Bath
49	Empowering undergraduate computer science students to shape generative AI research (Published: 15 July 2024)	Blog/Magazine/Social media post	Bobby Whyte, Raspberry PI Foundation
50	How can Generative AI Benefit Educators in Designing Assessments in Computer Science? (Published: Dec 2024)	Journal Article	Ling Wang and Shen Zhan
51	Imagining Computing Education Assessment after Generative AI (Published: 9 Jan 2024)	Journal Article	Stephen MacNeil, Scott Spurlock, and Ian Applebaum
52	Generative AI is changing how computer science is taught (Published: 25 Sep 2023)	Blog/Magazine/Social media post	Chris Stokel-Walker
53	Assessment in CS50 with AI: Leveraging Generative Artificial Intelligence for Personalized Student Evaluation (Published: 18 Feb 2025)	Conference Proceedings	Rongxin Liu, Benjamin Xu, Christopher Perez, Julianna Zhao, Yuliia Zhukovets, David J. Malan
54	Risk management strategy for generative AI in computing education: how to handle the strengths, weaknesses, opportunities, and threats? (Published: 11 Dec 2024)	Journal Article	Niklas Humble
55	Generative AI and Software Engineering Education (Published: 9 Sep 2024)	Blog/Magazine/Social media post	Ipek Ozkaya and Douglas Schmidt
56	Elisa Baniassad explores using ChatGPT to create Computer Science assessments and practice materials (Accessed: 8 Mar 2025)	Blog/Magazine/Social media post	Elisa Baniassad, The University of British Colombia
57	Integrating generative ai methods in computer science education: perspectives, strategies, and outcomes (Published: 3 July, 2024)	Conference Proceedings	M. Zimmermann, H. Janetzko, B. Haymond
58	Transforming Computer Science Education with Generative AI: My Journey (Published: 1 Jun 2024)	Blog/Magazine/Social media post	Harmeet Kaur

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60	Applying Generative Artificial Intelligence to Critiquing Science Assessments (Published: 30 Oct 2024)	Journal Article	Ha Nguyen, and Jake Hayward
61	Assessing Learning of Computer Programming Skills in the Age of Generative Artificial Intelligence (Published: May 2024)	Journal Article	Sara Ellen Wilson, and Matthew Nishimoto
62	Generative AI in Computer Science Education (Published: 14 Mar 2024)	Journal Article	Orit Hazzan
63	ChatGPT in Computer Science Curriculum Assessment: An analysis of Its Successes and Shortcomings (Published: 16 Apr 2023)	Conference Proceedings	B. Qureshi
64	Higher education assessment practice in the era of generative AI tools (Published: 1 April 2024)	Journal Article	Bayode Ogunleye et al
65	Generative AI in the Software Modeling Classroom: An Experience Report With ChatGPT and Unified Modeling Language (Published: Nov.-Dec. 2024)	Blog/Magazine/Social media post	Javier Cámara, Javier Troya, Julio Montes-Torres, and Francisco J. Jaime, IEEE Software
66	Can generative AI and ChatGPT outperform humans on cognitive-demanding problem-solving tasks in science? (Published: 7 Jan 2024)	Journal Article	X Zhai, M Nyaaba, and W Ma
67	Project based assessment in the era of generative AI challenges and opportunities (Published: Jun 2024)	Conference Proceedings	Naouel Boughattas, Wissal Neji, and Faten Ziadi
68	Engineered Prompts in ChatGPT for Educational Assessment in Software Engineering and Computer Science (Published: 26 Jan 2025)	Journal Article	Ayman Diyab, Russell Morris Frost, Benjamin David Fedoruk, andAhmad Diyab
69	Generative AI-Resistant Assessment Design (Accessed: 8 Mar 2025)	Guidance Document	Nadil Sundaraperuma/Dr. Naga Dasari, IIBIT/FedUni
70	Generative AI Is Changing Computer Science Education For The Better (Published: 6 Oct 2023)	Blog/Magazine/Social media post	Carla Jose
71	The Future of Exams: How Generative AI is Revolutionizing Assessments (Accessed: 8 Mar 2025)	Blog/Magazine/Social media post	Eklavvya
72	How to Teach Computer Programming in the Era of Generative AI: An Overview of the PRIMM Model (Published: 21 Jul 2023)	Blog/Magazine/Social media post	Mike McMillan
73	A New Kind of Innovation for an AI World (Published: 13 Dec 2024)	Blog/Magazine/Social media post	Turgay DEMİREL
74	Enhancing programming competency of Computer Science students by comparing student solutions with GenAI outputs (Accessed: 8 Mar 2025)	Guidance Document	Béibhinn Nic Ruairí and Dr. Jonathan Dukes
75	Generative Artificial Intelligence in Information Systems Education: Challenges, Consequences, and Responses (Published: 22 Jun 2023)	Journal Article	Craig Van Slyke, Richard D. Johnson and Jalal Sarabadani

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76	Evaluating the Integration of Generative AI in ICT Higher Education: Aligning University Practices with TEQSA Principles for Effective and Ethical Assessment (Published: 2024)	Research Report	Amara Atif, Meena Jha and Deborah Richards
77	The Promise and Pitfalls: A Literature Review of Generative Artificial Intelligence as a Learning Assistant in ICT Education (Published: 24 January 2025)	Journal Article	Ritesh Chugh et al
78	Exploration of the Application of ChatGPT in Information Technology Courses (Published: 12 Aug 2024)	Conference Proceedings	Mingjie Du; Yongfang Mao; Yanwen Wang; Fu Xie
79	Exploring the impact of using Chat-GPT on student learning outcomes in technology learning: The comprehensive experiment (Published: Oct 18, 2023)	Journal Article	Muhammad Hakiki et al
80	Considerations for adapting higher education technology courses for AI large language models: A critical review of the impact of ChatGPT (Published: Mar 2024)	Journal Article	Tayan, O., Hassan, A., Khankan, K., and Askool, S

Appendix 2

Order of Code in List	Nested Level	Code Name	Number of Snippets	Order of Code in List	Nested Level	Code Name	Number of Snippets
1	>	CS assessment Aspects	11	93	> >	Scaffolded Use of AI	15
2	> >	Automated Assessment Systems	19	94	> >	Program level strategies	29
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7	> >	Software Engineering	8	99	> >	Authentic assessments as a strategy	44
8	> >	Data Science & ML	9	100	> >	Continuous Assessments as a strategy	27
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10	> >	Motivate students	11	102	> >	Alignment—students' workload	11
11	> > >	self-efficacy	7	103	> >	Alignment-Differently-abled students	40
12	> > >	Upgrading	2	104	> >	Generalising by assessment types is dangerous	2
13	> >	Source Code Plagiarism	2	105	> >	discipline specific assessments	3
14	> >	Help Seeking preference	5	106	> >	Cohort specific assessments	2
15	> >	GenAI should be taught as a key topic	6	107	> >	Strategies—Broad list	34
16	> >	Entrepreneurship/ Apprenticeship assessments	1	108	> >	Shift the focus from grades to learning outcomes	13

Order of Code in List	Nested Level	Code Name	Number of Snippets	Order of Code in List	Nested Level	Code Name	Number of Snippets
17	> >	Networking/security assignments	3	109	> >	Critical Thinking as a strategy	50
18	> >	New skills in AI era	38	110	> >	Include GenAI to the assessment	25
19	> > >	Code inspection Skill	3	111	> >	Test your assessment with GenAI	17
20	> >	Fundamental Principals	13	112	>	GenAI Literacy	7
21	> >	Prompt Engineering	35	113	> >	Lifelong learning	16
22	> >	Programming skills	49	114	> > >	Setting Boundaries	3
23	> > >	Quality of the code	12	115	> >	Graduateness	32
24	> >	System scalability and Optimisation considerations	1	116	> >	Up-skilling Academics	52
25	> >	Algorithm selection and justification	9	117	> >	Support students to be GenAI literate	64
26	> >	Test Case Design	3	118	>	Ethics Privacy and Security Considerations	18
27	> >	CS assessment Strategies	34	119	> >	Professional Concerns	7
28	> > >	Self-evaluation of code	7	120	> >	Students' Anxiety and Emotional competence	26
29	> > >	PRIMM Model	1	121	> >	Assessment security and privacy	14
30	> > >	Real-time Coding Assessments	3	122	> >	Ethical, moral, and civic values	55
31	> > >	Less theory More Application	3	123	>	AI Usage in Assessments	25
32	> > >	From Correctness of Code to Understanding—In-Person Grading	4	124	> >	Communications with students	28
33	> > >	AI as Round the clock tutor	18	125	> >	Cite GenAI	7
34	> > >	In class activities	2	126	> >	Interdisciplinary learning with AI	6
35	> > >	Higher order thinking	24	127	> >	AI is Good at	27
36	> >	Debugging	5	128	> >	Difficulty for AI technologies Task Suggestion	17
37	> >	Assessing the ability to prompt an AI	2	129	> >	GenAI is bad at	39
38	> >	Problem decomposition	9	130	> >	providing feedback to students	24
39	> >	Programming Syntax	8	131	> > >	Using Gen AI to Provide Feedback to Students	8
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41	>	GenAI tools for learning and teaching	51	133	> >	Allow use with acknowledgement	26
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43	>	Cons of GenAI	20	135	> >	Gap—Current Situation with AI	11
44	> >	creativity	4	136	> >	Use AI Innovatively	28
45	> >	Hallucinations	11	137	> >	Test assessments with GenAI	11
46	>	Students' Opinions	11	138	>	Assessment Types	6

Order of Code in List	Nested Level	Code Name	Number of Snippets	Order of Code in List	Nested Level	Code Name	Number of Snippets
47	> >	Biasedness of evaluation from AI tools	10	139	> >	Disposable Assessments	2
48	> >	Student's AI usage patterns	15	140	> >	Observations	4
49	> >	usefulness of GenAI	15	141	> >	Portfolios	5
50	>	Assessment strategies	40	142	> >	Experiential Project based learning	7
51	> >	Interactive Tutorials	21	143	> >	Situational Context based assessments	6
52	> >	Use multi step problems	2	144	> > >	Simulations and Scenarios	13
53	> >	Avoid factual Recall	10	145	> > >	Role plays	4
54	> >	Choose examples from Global South	1	146	> >	Practicals (labs and tutorials)	8
55	> >	Reduce the volume of assessments	6	147	> >	Summative & Formative Assessments	10
56	> >	Contextual/personal assessments	19	148	> >	Group Projects	19
57	> >	Security at meaningful points across a program	5	149	> >	Personalised assessments	23
58	> >	Assessment tasks that uses AI	6	150	> >	Interviews or presentations	33
59	> >	Collaboration between AI and Peer Students	4	151	> >	Application based assessments	8
60	> >	Need to triangulate	1	152	> >	Frequent low stakes Assessments	12
61	> >	Allow AI tools for assessments	13	153	> >	Peer Assessments	11
62	> >	Submit annotated bibliography	7	154	> >	Problem Solving Assessments	4
63	> >	Marking criteria / Rubrics / Pass mark	16	155	> >	Real life situations and practical experience	18
64	> >	Allowing Multiple attempts	1	156	> >	Quizzes	13
65	> >	Length of assessment	5	157	> >	Essays	7
66	> >	Loopholes in assessment design	17	158	> >	Alternatives to Essay	15
67	> >	Controlled settings	9	159	> >	Learning Journals	11
68	> >	Using Subject matter expertise	2	160	> >	Examinations	24
69	> >	Flipped Assessment—solve with AI and assess outcome	2	161	> > >	Open Book exams	4
70	> >	Competency based grading	6	162	>	Policies, Procedures & Governance	8
71	> >	Cultural Diversity	2	163	> >	Risk Assessment and Planning	10
72	> >	Limiting to institutional subscription	3	164	> >	Innovation Speed	2
73	> >	Time Allocated for assessments	7	165	> >	Governance and Strategic Planning	18
74	> >	In Class Assessments	18	166	> >	Flexible assessment policies	6
75	> >	Types of assessments to avoid	7	167	> >	Policies for academic integrity	41

Order of Code in List	Nested Level	Code Name	Number of Snippets	Order of Code in List	Nested Level	Code Name	Number of Snippets
76	> >	In class concept maps	4	168	> >	Threat to academic Integrity	11
77	> >	Reflection via Peer Review	9	169	> >	Alignment—policy, & procedure	30
78	> >	Shift from assessing product to process	14	170	> > >	academic misconducts	30
79	> > >	Assess the Process	59	171	> > >	Special Considerations	3
80	> > >	Metacognitive Skills Development	12	172	> >	Alignment with TEQSA	1
81	> >	Behavioural and attitude changes	1	173	> >	Missing Regulations	3
82	> > >	Reduce motivation to cheat	3	174	>	AI Detection	37
83	> >	good assessment design	22	175	> >	Using GAI paraphrasing tools	1
84	> >	Feedback loop between teacher and student	19	176	> >	Supervision / invigilation	16
85	> >	Communication and confidence building	9	177	>	Frameworks proposed	18
86	> >	Returning Old School	15	178	> >	Activation Approach	1
87	> >	Reduce Risk Strategies	17	179	> >	Framework to assess Authentic Assessments	12
88	> > >	If AI is not allowed, must use mechanisms to detect	1	180	> >	Levels of AI Usage	22
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92	> >	Hurdles	1	184	>	Good practices	27
				185	>	Challenges	41

Abbreviations

ADM	Assessment design datrix
AQF	Australian Qualification Framework
GenAI	Generative artificial intelligence
GENESIS	GenAI-enabled strategic and instructional system for assessment
ICT	Information and communication technology
LLM	Large language models
TA	Thematic analysis

Supplementary Information

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Supplementary Material 1

Author contributions

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Data availability
The study is entirely based on secondary, web-based documentary data that are publicly available. All data analysed in this study are included in Appendix 1 of this published article. Appendix 1 provides detailed information on the documents used for analysis. No additional quantitative datasets were utilised.

Declarations

Ethics approval and consent to participate
This study is completely based on secondary web-based document data that are publicly available. Therefore, an ethics approval is not applicable.

Competing interests
There are no competing interests to declare in this study. The research is solely driven by academic interest, with no associated financial interests or other potential biases.

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